

NVIDIA ConnectX-6 InfiniBand/Ethernet Adapter Cards User Manual

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About This Manual

This User Manual describes NVIDIA® ConnectX®-6 InfiniBand/Ethernet adapter cards. It provides details as to the interfaces of the board, specifications, required software and firmware for operating the board, and relevant documentation.

Ordering Part Numbers

The table below provides the ordering part numbers (OPN) for the available ConnectX-6 InfiniBand/Ethernet adapter cards.

NVIDIA SKU	Legacy OPN	Marketing Description
900-9X6AF-0018-MT2	MCX653105A-HDAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **HDR IB** (200Gb/s) and 200GbE, **single**-port QSFP56, **PCIe3.0/4.0 x16**, tall bracket
900-9X6AF-0058-ST1	MCX653106A-HDAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **HDR IB** (200Gb/s) and 200GbE, **dual**-port QSFP56, **PCIe3.0/4.0 x16**, tall bracket

EOL'ed (End of Life) Ordering Part Numbers

NVIDIA SKU	Legacy OPN	Marketing Description
900-9X603-0016-DT0	MCX653105A-EFAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **100Gb/s** (HDR100, EDR IB and 100GbE), **single**-port QSFP56, **PCIe3.0/4.0 Socket Direct 2x8 in a row**, tall bracket
900-9X603-0056-DT0	MCX653106A-EFAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **100Gb/s** (HDR100, EDR IBand100GbE), **dual**-port QSFP56, **PCIe3.0/4.0 Socket Direct 2x8 in a row**, tall bracket
900-9X628-0016-ST0	MCX651105A-EDAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **100Gb/s** (HDR100, EDR IB and 100GbE, **single**-port QSFP56, **PCIe4.0 x8**, tall bracket
900-9X6AF-0016-ST1	MCX653105A-ECAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **100Gb/s** (HDR100, EDR IB and 100GbE), **single**-port QSFP56, **PCIe3.0/4.0 x16**, tall bracket
900-9X6AF-0056-MT1	MCX653106A-ECAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **100Gb/s** (HDR100, EDR IB and 100GbE), **dual**-port QSFP56, **PCIe3.0/4.0 x16**, tall bracket

NVIDIA SKU	Legacy OPN	Marketing Description
900-9X6B4-0058-DT0	MCX654106A-HCAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **HDR IB** (200Gb/s) and 200GbE, **dual**-port QSFP56, **Socket Direct 2x PCIe3.0/4.0x16**, tall bracket
900-9X6AF-0018-SS0	MCX653105A-HDAL	ConnectX®-6 InfiniBand/Ethernet adapter card, **HDR IB** (200Gb/s) and 200GbE, **single**-port QSFP56, **PCIe3.0/4.0 x16**, **cold plate for liquid-cooled Intel® Server System D50TNP platforms**, tall bracket, ROHS R6
900-9X6AF-0058-SS0	MCX653106A-HDAL	ConnectX®-6 InfiniBand/Ethernet adapter card, **HDR IB** (200Gb/s) and 200GbE, **dual**-port QSFP56, **PCIe3.0/4.0 x16**, **cold plate for liquid-cooled Intel® Server System D50TNP platforms**, tall bracket, ROHS R6
900-9X0BC-001H-ST1	MCX683105AN-HDAT	ConnectX®-6 DE adapter card, **HDR IB** (200Gb/s), **single**-port QSFP, **PCIe4.0 x16,** no crypto, tall bracket
900-9X6B4-0056-DT0	MCX654106A-ECAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **100Gb/s** (HDR100, EDR InfiniBand and 100GbE), **dual**-port QSFP56, **Socket Direct 2x PCIe 3.0/4.0** x16, tall bracket
900-9X6B4-0018-DT2	MCX654105A-HCAT	ConnectX®-6 InfiniBand/Ethernet adapter card, **HDR IB** (200Gb/s) and 200GbE, **single**-port QSFP56, **Socket Direct 2x PCIe3.0/4.0x16**, tall bracket

Intended Audience

This manual is intended for the installer and user of these cards.

The manual assumes basic familiarity with InfiniBand and Ethernet network and architecture specifications.

Technical Support

Customers who purchased NVIDIA products directly from NVIDIA are invited to contact us through the following methods:

- URL: https://www.nvidia.com > Support
- E-mail: enterprisesupport@nvidia.com

Customers who purchased NVIDIA M-1 Global Support Services, please see your contract for details regarding Technical Support.

Customers who purchased NVIDIA products through a NVIDIA approved reseller should first seek assistance through their reseller.

Related Documentation

MLNX_OFED for Linux User Manual and Release Notes	User Manual describing OFED features, performance, band diagnostic, tools content, and configuration. See MLNX_OFED for Linux Documentation.
WinOF-2 for Windows User Manual and Release Notes	User Manual describing WinOF-2 features, performance, Ethernet diagnostic, tools content, and configuration. See WinOF-2 for Windows Documentation.
NVIDIA VMware for Ethernet User Manual	User Manual and release notes describing the various components of the NVIDIA ConnectX® NATIVE ESXi stack. See VMware® ESXi Drivers Documentation.
NVIDIA Firmware Utility (mlxup) User Manual and Release Notes	NVIDIA firmware update and query utility used to update the firmware. Refer to Firmware Utility (mlxup) Documentation.
NVIDIA Firmware Tools (MFT) User Manual	User Manual describing the set of MFT firmware management tools for a single node. See MFT User Manual.
InfiniBand Architecture Specification Release 1.2.1, Vol 2 - Release 1.3	InfiniBand Specifications
IEEE Std 802.3 Specification	IEEE Ethernet Specifications
PCI Express Specifications	Industry Standard PCI Express Base and Card Electromechanical Specifications. Refer to PCI-SIG Specifications.

LinkX Interconnect Solutions	LinkX InfiniBand cables and transceivers are designed to maximize the performance of High-Performance Computing networks, requiring high-bandwidth, low-latency connections between compute nodes and switch nodes. NVIDIA offers one of the industry's broadest portfolio of QDR/FDR10 (40Gb/s), FDR (56Gb/s), EDR/HDR100 (100Gb/s), HDR (200Gb/s) and NDR (400Gb/s) cables, including Direct Attach Copper cables (DACs), copper splitter cables, Active Optical Cables (AOCs) and transceivers in a wide range of lengths from 0.5m to 10km. In addition to meeting IBTA standards, NVIDIA tests every product in an end-to-end environment ensuring a Bit Error Rate of less than 1E-15 . Read more at LinkX Cables and Transceivers.

Document Conventions

When discussing memory sizes, MB and MBytes are used in this document to mean size in MegaBytes. The use of Mb or Mbits (small b) indicates the size in MegaBits. In this document, PCIe is used to mean PCI Express.

Revision History

A list of the changes made to this document are provided in Document Revision History.

Introduction

Product Overview

This is the user guide for InfiniBand/Ethernet adapter cards based on the ConnectX-6 integrated circuit device. ConnectX-6 connectivity provides the highest performing low latency and most flexible interconnect solution for PCI Express Gen 3.0/4.0 servers used in enterprise datacenters and high-performance computing environments.

ConnectX-6 Virtual Protocol Interconnect® adapter cards provide up to two ports of 200Gb/s for InfiniBand and Ethernet connectivity, sub-600ns latency and 200 million messages per second, enabling the highest performance and most flexible solution for the most demanding High-Performance Computing (HPC), storage, and datacenter applications.

ConnectX-6 is a groundbreaking addition to the NVIDIA ConnectX series of industry-leading adapter cards. In addition to all the existing innovative features of past ConnectX versions, ConnectX-6 offers a number of enhancements that further improve the performance and scalability of datacenter applications. In addition, specific PCIe stand-up cards are available with a cold plate for insertion into liquid-cooled Intel® Server System D50TNP platforms.

ConnectX-6 is available in two form factors: low-profile stand-up PCIe and Open Compute Project (OCP) Spec 3.0 cards with QSFP connectors. Single-port, HDR, stand-up PCIe adapters are available based on either ConnectX-6 or ConnectX-6 DE (ConnectX-6 Dx enhanced for HPC applications).

Note

Make sure to use a PCIe slot that is capable of supplying the required power and airflow to the ConnectX-6 as stated in Specifications.

Configuration	OPN	Marketing Description
ConnectX-6 PCIe x8 Card	MCX651105A-EDAT	ConnectX-6 InfiniBand/Ethernet adapter card, 100Gb/s (HDR100, EDR IB and 100GbE, single-port QSFP56, PCIe4.0 x8, tall bracket
ConnectX-6 PCIe x16 Card	MCX653105A-HDAT	ConnectX-6 InfiniBand/Ethernet adapter card, HDR IB (200Gb/s) and 200GbE, single-port QSFP56, PCIe4.0 x16, tall bracket
	MCX653106A-HDAT	ConnectX-6 InfiniBand/Ethernet adapter card, HDR IB (200Gb/s) and 200GbE, dual-port QSFP56, PCIe3.0/4.0 x16, tall bracket
	MCX653105A-ECAT	ConnectX-6 InfiniBand/Ethernet adapter card, 100Gb/s (HDR100, EDR IB and 100GbE), single-port QSFP56, PCIe3.0/4.0 x16, tall bracket
	MCX653106A-ECAT	ConnectX-6 InfiniBand/Ethernet adapter card, 100Gb/s (HDR100, EDR IB and 100GbE), dual-port QSFP56, PCIe3.0/4.0 x16, tall bracket
ConnectX-6 DE PCIe x16 Card	MCX683105AN-HDAT	ConnectX-6 DE InfiniBand adapter card, HDR, single-port QSFP, PCIe 3.0/4.0 x16, No Crypto, Tall Bracket
ConnectX-6 PCIe x16 Cards for liquid-cooled Intel® Server System D50TNP platforms	MCX653105A-HDAL	ConnectX-6 InfiniBand/Ethernet adapter card, HDR IB (200Gb/s) and 200GbE, single-port QSFP56, PCIe4.0 x16, cold plate for liquid-cooled Intel® Server System D50TNP platforms, tall bracket, ROHS R6
	MCX653106A-HDAL	ConnectX-6 InfiniBand/Ethernet adapter card, HDR IB (200Gb/s) and 200GbE, dual-port QSFP56, PCIe4.0 x16, cold plate for liquid-cooled Intel® Server System D50TNP platforms, tall bracket, ROHS R6

Configuration	OPN	Marketing Description
ConnectX-6 Dual-slot Socket Direct Cards (2x PCIe x16)	MCX654105A-HCAT	ConnectX-6 InfiniBand/Ethernet adapter card kit, HDR IB (200Gb/s) and 200GbE, single-port QSFP56, Socket Direct 2x PCIe3.0 x16, tall brackets
	MCX654106A-HCAT	ConnectX-6 InfiniBand/Ethernet adapter card, HDR IB (200Gb/s) and 200GbE, dual-port QSFP56, Socket Direct 2x PCIe3.0/4.0x16, tall bracket
	MCX654106A-ECAT	ConnectX-6 InfiniBand/Ethernet adapter card, 100Gb/s (HDR100, EDR InfiniBand and 100GbE), dual-port QSFP56, Socket Direct 2x PCIe3.0/4.0 x16, tall bracket
ConnectX-6 Single-slot Socket Direct Cards (2x PCIe x8 in a row)	MCX653105A-EFAT	ConnectX-6 InfiniBand/Ethernet adapter card, 100Gb/s (HDR100, EDR IB and 100GbE), single-port QSFP56, PCIe3.0/4.0 Socket Direct 2x8 in a row, tall bracket
	MCX653106A-EFAT	ConnectX-6 InfiniBand/Ethernet adapter card, 100Gb/s (HDR100, EDR IBand100GbE), dual-port QSFP56, PCIe3.0/4.0 Socket Direct 2x8 in a row, tall bracket

ConnectX-6 PCIe x8 Card

ConnectX-6 with a single PCIe x8 slot can support a bandwidth of up to 100Gb/s in a PCIe Gen 4.0 slot.

Part Number	MCX651105A-EDAT
Form Factor/Dimensions	PCIe Half Height, Half Length / 167.65mm x 68.90mm
Data Transmission Rate	**Ethernet**: 10/25/40/50/100 Gb/s **InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100
Network Connector Type	Single-port QSFP56
PCIe x8 through Edge Connector	PCIe Gen 3.0 / 4.0 SERDES @ 8.0GT/s / 16.0GT/s
RoHS	RoHS Compliant

Part Number	MCX651105A-EDAT
Adapter IC Part Number	MT28908A0-XCCF-HVM

ConnectX-6 PCIe x16 Card

ConnectX-6 with a single PCIe x16 slot can support a bandwidth of up to 100Gb/s in a PCIe Gen 3.0 slot, or up to 200Gb/s in a PCIe Gen 4.0 slot. This form-factor is available also for Intel® Server System D50TNP Platforms where an Intel liquid-cooled cold plate is used for adapter cooling mechanism.

Part Number	MCX653105A-ECAT	MCX653106A-ECAT	MCX653105A-HDAT	MCX653106A-HDAT
Form Factor/Dimensions	PCIe Half Height, Half Length / 167.65mm x 68.90mm			
Data Transmission Rate	**Ethernet**: 10/25/40/50/100 Gb/s **InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100		**Ethernet**: 10/25/40/50/100/200 Gb/s **InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100, HDR	
Network Connector Type	Single-port QSFP56	Dual-port QSFP56	Single-port QSFP56	Dual-port QSFP56
PCIe x16 through Edge Connector	PCIe Gen 3.0 / 4.0 SERDES @ 8.0GT/s / 16.0GT/s			
RoHS	RoHS Compliant			
Adapter IC Part Number	MT28908A0-XCCF-HVM			

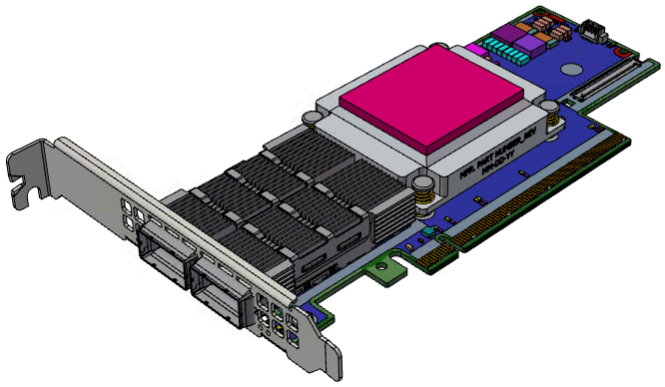
ConnectX-6 DE PCIe x16 Card

ConnectX-6 DE (ConnectX-6 Dx enhanced for HPC applications) with a single PCIe x16 slot can support a bandwidth of up to 100Gb/s in a PCIe Gen 3.0 slot, or up to 200Gb/s in a PCIe Gen 4.0 slot.

Part Number	MCX683105AN-HDAT
Form Factor/Dimensions	PCIe Half Height, Half Length / 167.65mm x 68.90mm
Data Transmission Rate	**InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100, HDR
Network Connector Type	Single-port QSFP56
PCIe x16 through Edge Connector	PCIe Gen 3.0 / 4.0 SERDES @ 8.0GT/s / 16.0GT/s
RoHS	RoHS Compliant
Adapter IC Part Number	MT28924A0-NCCF-VE

ConnectX-6 for Liquid-Cooled Intel® Server System D50TNP Platforms

The below cards are available with a cold plate for insertion into liquid-cooled Intel® Server System D50TNP platforms.



Part Number	MCX653105A-HDAL	MCX653106A-HDAL
Form Factor/Dimensions	PCIe Half Height, Half Length / 167.65mm x 68.90mm	
Data Transmission Rate	**Ethernet**: 10/25/40/50/100/200 Gb/s **InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100, HDR	
Network Connector Type	Single-port QSFP56	Dual-port QSFP56

Part Number	MCX653105A-HDAL	MCX653106A-HDAL
PCIe x16 through Edge Connector	PCIe Gen 3.0 / 4.0 SERDES @ 8.0GT/s / 16.0GT/s	
RoHS	RoHS Compliant	
Adapter IC Part Number	MT28908A0-XCCF-HVM	

ConnectX-6 Socket Direct™ Cards

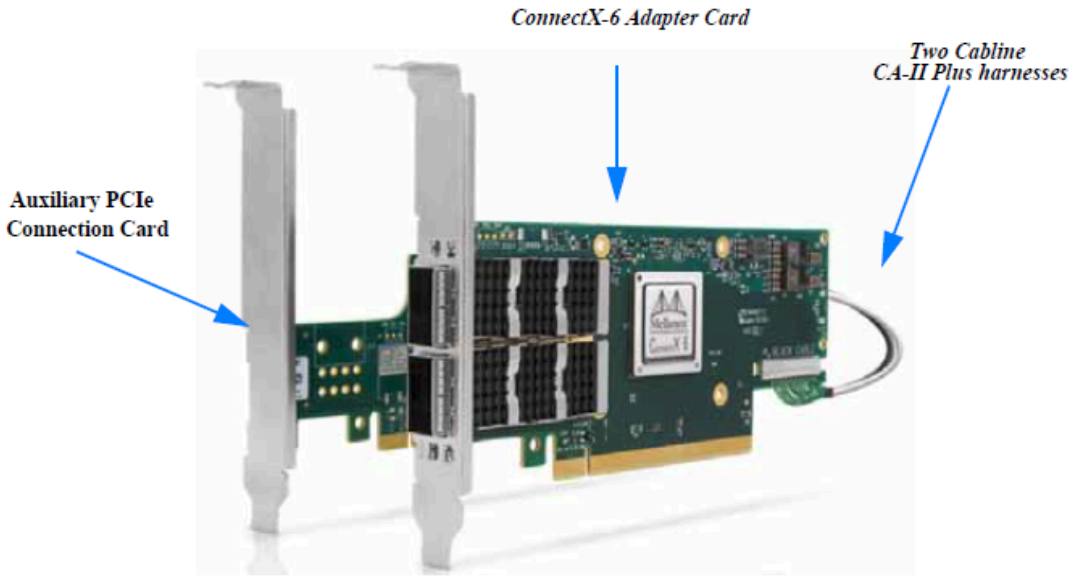
The Socket Direct technology offers improved performance to dual-socket servers by enabling direct access from each CPU in a dual-socket server to the network through its dedicated PCIe interface.

Please note that ConnectX-6 Socket Direct cards do not support Multi-Host functionality (i.e. connectivity to two independent CPUs). For ConnectX-6 Socket Direct card with Multi-Host functionality, please contact NVIDIA.

ConnectX-6 Socket Direct cards are available in two configurations: **Dual-slot Configuration** (2x PCIe x16) and **Single-slot Configuration** (2x PCIe x8).

ConnectX-6 Dual-slot Socket Direct Cards (2x PCIe x16)

In order to obtain 200Gb/s speed, NVIDIA offers ConnectX-6 Socket Direct that enable 200Gb/s connectivity also for servers with PCIe Gen 3.0 capability. The adapter's 32-lane PCIe bus is split into two 16-lane buses, with one bus accessible through a PCIe x16 edge connector and the other bus through an x16 Auxiliary PCIe Connection card. The two cards should be installed into two PCIe x16 slots and connected using two Cabline SA-II Plus harnesses, as shown in the below figure.



Part Number	MCX654105A-HCAT	MCX654106A-HCAT	MCX654106A-ECAT
Form Factor/Dimensions	Adapter Card: PCIe Half Height, Half Length / 167.65mm x 68.90mm Auxiliary PCIe Connection Card: 5.09 in. x 2.32 in. (129.30mm x 59.00mm)Two 35cm Cabline CA-II Plus harnesses		
Data Transmission Rate	**Ethernet**: 10/25/40/50/100/200 Gb/s **InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100, HDR		**Ethernet**: 10/25/40/50/100 Gb/s **InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100
Network Connector Type	Single-port QSFP56	Dual-port QSFP56	
PCIe x16 through Edge Connector	PCIe Gen 3.0 / 4.0SERDES@ 8.0GT/s / 16.0GT/s		
PCIe x16 through Auxiliary Card	PCIe Gen 3.0SERDES@ 8.0GT/s		
RoHS	RoHS Compliant		
Adapter IC Part Number	MT28908A0-XCCF-HVM		

ConnectX-6 Single-slot Socket Direct Cards (2x PCIe x8 in a row)

The PCIe x16 interface comprises two PCIe x8 in a row, such that each of the PCIe x8 lanes can be connected to a dedicated CPU in a dual-socket server. In such a configuration, Socket Direct brings lower latency and lower CPU utilization as the direct connection from each CPU to the network means the interconnect can bypass a QPI (UPI) and the other CPU, optimizing performance and improving latency. CPU utilization is improved as each CPU handles only its own traffic and not traffic from the other CPU.

A system with a custom PCI Express x16 slot that includes special signals is required for installing the card. Please refer to PCI Express Pinouts Description for Single-Slot Socket Direct Card for pinout definitions.

Part Number	MCX653105A-EFAT	MCX653106A-EFAT
Form Factor/Dimensions	PCIe Half Height, Half Length / 167.65mm x 68.90mm	
Data Transmission Rate	**Ethernet**: 10/25/40/50/100 Gb/s **InfiniBand**: SDR, DDR, QDR, FDR, EDR, HDR100	
Network Connector Type	Single-port QSFP56	Dual-port QSFP56
PCIe x16 through Edge Connector	PCIe Gen 3.0 / 4.0 SERDES @ 8.0GT/s / 16.0GT/s Socket Direct 2x8 in a row	
RoHS	RoHS Compliant	
Adapter IC Part Number	MT28908A0-XCCF-HVM	

Package Contents

ConnectX-6 PCIe x8/x16 Adapter Cards

> **Note**
>
> Applies to MCX651105A-EDAT, MCX653105A-ECAT, MCX653106A-ECAT, MCX653105A-HDAT, MCX653106A-HDAT, MCX653105A-EFAT, MCX653106A-EFAT, and MCX683105AN-HDAT .

Category	Qty	Item
Cards	1	ConnectX-6 adapter card
Accessories	1	Adapter card short bracket
	1	Adapter card tall bracket (shipped assembled on the card)

ConnectX-6 PCIe x16 Adapter Card for liquid-cooled Intel® Server System D50TNP Platforms

Note

Applies to MCX653105A-HDAL and MCX653106A-HDAL.

Category	Qty	Item
Cards	1	ConnectX-6 adapter card
Accessories	1	Adapter card short bracket
	1	Adapter card tall bracket (shipped assembled on the card)
	1	Accessory Kit with two 2 TIMs (MEB000386)

ConnectX-6 Socket Direct Cards (2x PCIe x16)

Note

Applies to MCX654105A-HCAT, MCX654106A-HCAT and MCX654106A-ECAT.

Category	Qty.	Item
Cards	1	ConnectX-6 adapter card
	1	PCIe Auxiliary Card
Harnesses	1	35cm Cabline CA-II Plus harness (white)
	1	35cm Cabline CA-II Plus harness (black)
Accessories	2	Retention Clip for Cablline harness (optional accessory)
	1	Adapter card short bracket
	1	Adapter card tall bracket (shipped assembled on the card)
	1	PCIe Auxiliary card short bracket
	1	PCIe Auxiliary card tall bracket (shipped assembled on the card)

Features and Benefits

Note

Make sure to use a PCIe slot that is capable of supplying the required power and airflow to the ConnectX-6 cards as stated in Specifications.

PCI Express (PCIe)	Uses the following PCIe interfaces: • PCIe x8/x16 configurations: PCIe Gen 3.0 (8GT/s) and Gen 4.0 (16GT/s) through an x8/x16 edge connector. • 2x PCIe x16 configurations: PCIe Gen 3.0/4.0 SERDES @ 8.0/16.0 GT/s through Edge ConnectorPCIe Gen 3.0 SERDES @ 8.0GT/s through PCIe Auxiliary Connection Card

200Gb/s InfiniBand/Ethernet Adapter	ConnectX-6 offers the highest throughput InfiniBand/Ethernet adapter, supporting HDR 200b/s InfiniBand and 200Gb/s Ethernet and enabling any standard networking, clustering, or storage to operate seamlessly over any converged network leveraging a consolidated software stack.
InfiniBand Architecture Specification v1.3 compliant	ConnectX-6 delivers low latency, high bandwidth, and computing efficiency for performance-driven server and storage clustering applications. ConnectX-6 is InfiniBand Architecture Specification v1.3 compliant.
Up to 200 Gigabit Ethernet	NVIDIA adapters comply with the following IEEE 802.3 standards: 200GbE / 100GbE / 50GbE / 40GbE / 25GbE / 10GbE / 1GbE- IEEE 802.3bj, 802.3bm 100 Gigabit Ethernet- IEEE 802.3by, Ethernet Consortium25, 50 Gigabit Ethernet, supporting all FEC modes- IEEE 802.3ba 40 Gigabit Ethernet- IEEE 802.3by 25 Gigabit Ethernet- IEEE 802.3ae 10 Gigabit Ethernet- IEEE 802.3ap based auto-negotiation and KR startup- IEEE 802.3ad, 802.1AX Link Aggregation- IEEE 802.1Q, 802.1P VLAN tags and priority- IEEE 802.1Qau (QCN)- Congestion Notification- IEEE 802.1Qaz (ETS)- IEEE 802.1Qbb (PFC)- IEEE 802.1Qbg- IEEE 1588v2- Jumbo frame support (9.6KB)
InfiniBand HDR100	A standard InfiniBand data rate, where each lane of a 2X port runs a bit rate of 53.125Gb/s with a 64b/66b encoding, resulting in an effective bandwidth of 100Gb/s.
InfiniBand HDR	A standard InfiniBand data rate, where each lane of a 4X port runs a bit rate of 53.125Gb/s with a 64b/66b encoding, resulting in an effective bandwidth of 200Gb/s.
Memory Components	• SPI Quad - includes 256Mbit SPI Quad Flash device (MX25L25645GXDI-08G device by Macronix) • FRU EEPROM - Stores the parameters and personality of the card. The EEPROM capacity is 128Kbit. FRU I2C address is (0x50) and is accessible through the PCIe SMBus. (Note: A ddress 0x58 is reserved.)

Overlay Networks	In order to better scale their networks, datacenter operators often create overlay networks that carry traffic from individual virtual machines over logical tunnels in encapsulated formats such as NVGRE and VXLAN. While this solves network scalability issues, it hides the TCP packet from the hardware offloading engines, placing higher loads on the host CPU. ConnectX-6 effectively addresses this by providing advanced NVGRE and VXLAN hardware offloading engines that encapsulate and de-capsulate the overlay protocol.
RDMA and RDMA over Converged Ethernet (RoCE)	ConnectX-6, utilizing IBTA RDMA (Remote Data Memory Access) and RoCE (RDMA over Converged Ethernet) technology, delivers low-latency and high-performance over InfiniBand and Ethernet networks. Leveraging datacenter bridging (DCB) capabilities as well as ConnectX-6 advanced congestion control hardware mechanisms, RoCE provides efficient low-latency RDMA services over Layer 2 and Layer 3 networks.
NVIDIA PeerDirect™	PeerDirect™ communication provides high efficiency RDMA access by eliminating unnecessary internal data copies between components on the PCIe bus (for example, from GPU to CPU), and therefore significantly reduces application run time. ConnectX-6 advanced acceleration technology enables higher cluster efficiency and scalability to tens of thousands of nodes.
CPU Offload	Adapter functionality enables reduced CPU overhead leaving more CPU resources available for computation tasks. Open vSwitch (OVS) offload using ASAP$^{2(TM)}$• Flexible match-action flow tables• Tunneling encapsulation/decapsulation
Quality of Service (QoS)	Support for port-based Quality of Service enabling various application requirements for latency and SLA.
Hardware-based I/O Virtualization	ConnectX-6 provides dedicated adapter resources and guaranteed isolation and protection for virtual machines within the server.
Storage Acceleration	A consolidated compute and storage network achieves significant cost-performance advantages over multi-fabric networks. Standard block and file access protocols can leverage: • RDMA for high-performance storage access • NVMe over Fabric offloads for target machine • Erasure Coding • T10-DIF Signature Handover

SR-IOV	ConnectX-6 SR-IOV technology provides dedicated adapter resources and guaranteed isolation and protection for virtual machines (VM) within the server.
High-Performance Accelerations	• Tag Matching and Rendezvous Offloads • Adaptive Routing on Reliable Transport • Burst Buffer Offloads for Background Checkpointing

Operating Systems/Distributions

Note

ConnectX-6 Socket Direct cards 2x PCIe x16 (OPNs: MCX654105A-HCAT, MCX654106A-HCAT and MCX654106A-ECAT) are not supported in Windows and WinOF-2.

- OpenFabrics Enterprise Distribution (OFED)
- RHEL/CentOS
- Windows
- FreeBSD
- VMware
- OpenFabrics Enterprise Distribution (OFED)
- OpenFabrics Windows Distribution (WinOF-2)

Connectivity

- Interoperable with 1/10/25/40/50/100/200 Gb/s InfiniBand and Ethernet switches
- Passive copper cable with ESD protection
- Powered connectors for optical and active cable support

Manageability

ConnectX-6 technology maintains support for manageability through a BMC. ConnectX-6 PCIe stand-up adapter can be connected to a BMC using MCTP over SMBus or MCTP over PCIe protocols as if it is a standard NVIDIA PCIe stand-up adapter. For configuring the adapter for the specific manageability solution in use by the server, please contact NVIDIA Support.

Interfaces

InfiniBand Interface

The network ports of the ConnectX®-6 adapter cards are compliant with the *InfiniBand Architecture Specification, Release 1.3*. InfiniBand traffic is transmitted through the cards' QSFP56 connectors.

Ethernet Interfaces

> **Note**
>
> The adapter card includes special circuits to protect from ESD shocks to the card/server when plugging copper cables.

The network ports of the ConnectX-6 adapter card are compliant with the IEEE 802.3 Ethernet standards listed in Features and Benefits. Ethernet traffic is transmitted through the QSFP56/QSFP connectors on the adapter card.

PCI Express Interface

ConnectX®-6 adapter cards support PCI Express Gen 3.0/4.0 (1.1 and 2.0 compatible) through x8/x16 edge connectors. The device can be either a master initiating the PCI Express bus operations or a subordinate responding to PCI bus operations.

The following lists PCIe interface features:

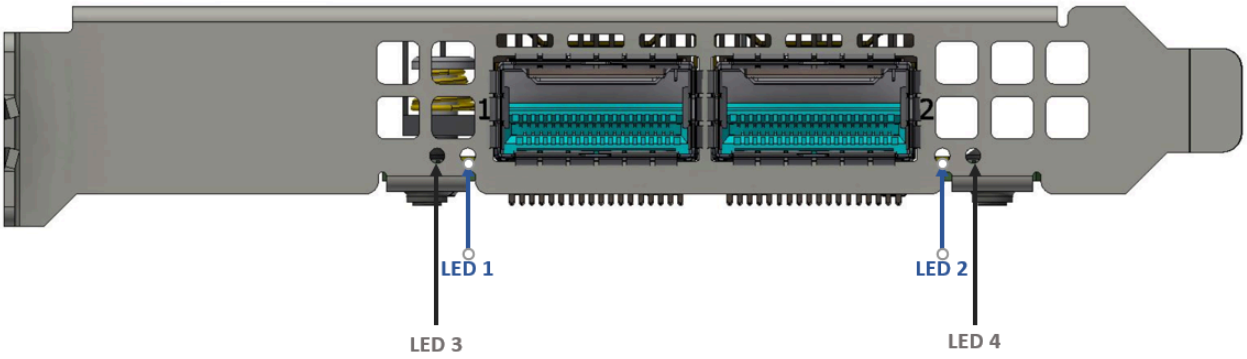
- PCIe Gen 3.0 and 4.0 compliant, 2.0 and 1.1 compatible
- 2.5, 5.0, 8.0, or 16.0 GT/s link rate x16/x32
- Auto-negotiates to x32, x16, x8, x4, x2, or x1
- Support for MSI/MSI-X mechanisms

LED Interface

> **Note**
>
> The adapter card includes special circuits to protect from ESD shocks to the card/server when plugging copper cables.

There are two I/O LEDs per port:

- LED 1 and 2: Bi-color I/O LED which indicates link status. LED behavior is described below for Ethernet and InfiniBand port configurations.
- LED 3 and 4: Reserved for future use.



LED1 and LED2 Link Status Indications - Ethernet Protocol:

LED Color and State	Description
Off	A link has not been established
Beacon command for locating the adapter card	1Hz blinking Yellow

<table>
<tr><td>Error</td><td>4Hz blinking Yellow Indicates an error with the link. The error can be one of the following:

<table><tr><th>Error Type</th><th>Description</th><th>LED Behavior</th></tr><tr><td>I²C</td><td>I²C access to the networking ports fails</td><td>Blinks until error is fixed</td></tr><tr><td>Over-current</td><td>Over-current condition of the networking ports</td><td>Blinks until error is fixed</td></tr></table></td></tr>
<tr><td>Solid green</td><td>Indicates a valid link with no active traffic</td></tr>
<tr><td>Blinking green</td><td>Indicates a valid link with active traffic</td></tr>
</table>

LED1 and LED2 Link Status Indications - InfiniBand Protocol:

<table>
<tr><th>LED Color and State</th><th>Description</th></tr>
<tr><td>Off</td><td>A link has not been established</td></tr>
<tr><td>Beacon command for locating the adapter card</td><td>1Hz blinking Yellow</td></tr>
<tr><td>Error</td><td>4Hz blinking Yellow Indicates an error with the link. The error can be one of the following:

<table><tr><th>Error Type</th><th>Description</th><th>LED Behavior</th></tr><tr><td>I²C</td><td>I²C access to the networking ports fails</td><td>Blinks until error is fixed</td></tr><tr><td>Over-current</td><td>Over-current condition of the networking ports</td><td>Blinks until error is fixed</td></tr></table></td></tr>
<tr><td>Solid amber</td><td>Indicates an active link</td></tr>
<tr><td>Solid green</td><td>Indicates a valid (data activity) link with no active traffic</td></tr>
<tr><td>Blinking green</td><td>Indicates a valid link with active traffic</td></tr>
</table>

Heatsink Interface

The heatsink is attached to the ConnectX-6 IC to dissipate the heat from the ConnectX-6 IC. It is attached either by using four spring-loaded push pins that insert into four mounting holes or by screws.

ConnectX-6 IC has a thermal shutdown safety mechanism that automatically shuts down the ConnectX-6 card in cases of high-temperature events, improper thermal coupling or heatsink removal.

For the required airflow (LFM) per OPN, please refer to Specifications.

For MCX653105A-HDAL and MCX653106A-HDAL cards, the heatsink is compatible with a cold plate for liquid-cooled Intel® Server System D50TNP platforms only.

SMBus Interface

ConnectX-6 technology maintains support for manageability through a BMC. ConnectX-6 PCIe stand-up adapter can be connected to a BMC using MCTP over SMBus protocol as if it is a standard NVIDIA PCIe stand-up adapter. For configuring the adapter for the specific manageability solution in use by the server, please contact NVIDIA Support.

Voltage Regulators

The voltage regulator power is derived from the PCI Express edge connector 12V supply pins. These voltage supply pins feed on-board regulators that provide the necessary power to the various components on the card.

Hardware Installation

Installation and initialization of ConnectX-6 adapter cards require attention to the mechanical attributes, power specification, and precautions for electronic equipment.

Safety Warnings

Info

Safety warnings are provided here in the English language. For safety warnings in other languages, refer to the Adapter Installation Safety Instructions document available on nvidia.com.

Please observe all safety warnings to avoid injury and prevent damage to system components.

Note that not all warnings are relevant to all models.

	**General Installation Instructions** Read all installation instructions before connecting the equipment to the power source.
	**Jewelry Removal Warning** Before you install or remove equipment that is connected to power lines, remove jewelry such as bracelets, necklaces, rings, watches, and so on. Metal objects heat up when connected to power and ground and can meltdown, causing serious burns and/or welding the metal object to the terminals.
	**Over-temperature** This equipment should not be operated in an area with an ambient temperature exceeding the maximum recommended: 55°C (131°F). An airflow of 200LFM at this maximum ambient temperature is required for HCA cards and NICs. To guarantee proper airflow, allow at least 8cm (3 inches) of clearance around the ventilation openings.

	**During Lightning - Electrical Hazard** During periods of lightning activity, do not work on the equipment or connect or disconnect cables.
	**Copper Cable Connecting/Disconnecting** Some copper cables are heavy and not flexible, as such, they should be carefully attached to or detached from the connectors. Refer to the cable manufacturer for special warnings and instructions.
	**Equipment Installation** This equipment should be installed, replaced, or serviced only by trained and qualified personnel.
	**Equipment Disposal** The disposal of this equipment should be in accordance to all national laws and regulations.
	**Local and National Electrical Codes** This equipment should be installed in compliance with local and national electrical codes.
	**Hazardous Radiation Exposure** • Caution – Use of controls or adjustment or performance of procedures other than those specified herein may result in hazardous radiation exposure.For products with optical ports. • CLASS 1 LASER PRODUCT and reference to the most recent laser standards: IEC 60 825-1:1993 + A1:1997 + A2:2001 and EN 60825-1:1994+A1:1996+ A2:20

Installation Procedure Overview

The installation procedure of ConnectX-6 adapter cards involves the following steps:

Step	Procedure	Direct Link
1	Check the system's hardware and software requirements.	System Requirements
2	Pay attention to the airflow consideration within the host system	Airflow Requirements
3	Follow the safety precautions	Safety Precautions
4	Unpack the package	Unpack the package

Step	Procedure	Direct Link
5	Follow the pre-installation checklist	Pre-Installation Checklist
6	(Optional) Replace the full-height mounting bracket with the supplied short bracket	Bracket Replacement Instructions
7	Install the ConnectX-6 PCIe x8/x16 adapter card in the system	ConnectX-6 PCIe x8/x16 Adapter Cards Installation Instructions
	Install the ConnectX-6 2x PCIe x16 Socket Direct adapter card in the system	Socket Direct (2x PCIe x16) Cards Installation Instructions
	Install the ConnectX-6 card for Intel Liquid-cooled platforms	Cards for Intel Liquid-Cooled Platforms Installation Instructions
8	Connect cables or modules to the card	Cables and Modules
9	Identify ConnectX-6 in the system	Identifying Your Card

System Requirements

Hardware Requirements

> ⚠ **Warning**
>
> Unless otherwise specified, NVIDIA products are designed to work in an environmentally controlled data center with low levels of gaseous and dust (particulate) contamination.
>
> The operating environment should meet severity level G1 as per ISA 71.04 for gaseous contamination and ISO 14644-1 class 8 for cleanliness level.

⚠ Warning

For proper operation and performance, please make sure to use a PCIe slot with a corresponding bus width and that can supply sufficient power to your card. Refer to the Specifications section of the manual for more power requirements.

⚠ Warning

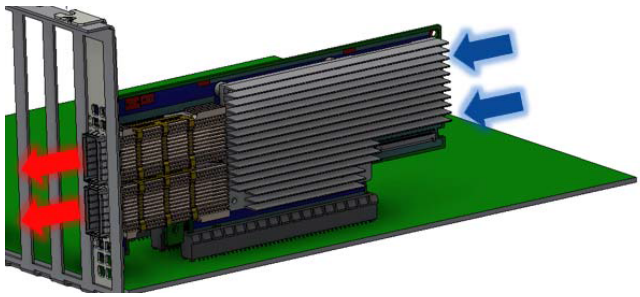
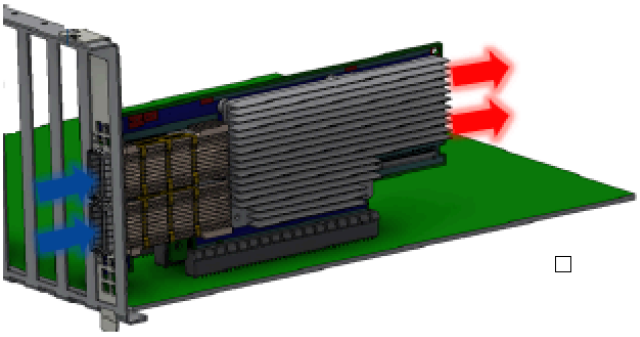
Please make sure to install the ConnectX-6 cards in a PCIe slot that is capable of supplying the required power as stated in Specifications.

ConnectX-6 Configuration	Hardware Requirements
PCIe x8/x16	A system with a PCI Express x8/x16 slot is required for installing the card.
Cards for liquid-cooled Intel® Server System D50TNP platforms	Intel® Server System D50TNP Platform with an available PCI Express x16 slot is required for installing the card.
Socket Direct 2x PCIe x8 in a row (single slot)	A system with a custom PCI Express x16 slot (four special pins) is required for installing the card. Please refer to PCI Express Pinouts Description for Single-Slot Socket Direct Card for pinout definitions.
Socket Direct 2x PCIe x16 (dual slots)	A system with two PCIe x16 slots is required for installing the cards.

Airflow Requirements

ConnectX-6 adapter cards are offered with two airflow patterns: from the heatsink to the network ports, and vice versa, as shown below.

Please refer to the Specifications section for airflow numbers for each specific card model.

Airflow from the heatsink to the network ports	Airflow from the network ports to the heatsink
	

> **Warning**
>
> All cards in the system should be planned with the same airflow direction.

Software Requirements

- See Operating Systems/Distributions section under the Introduction section.
- Software Stacks - NVIDIA OpenFabric software package MLNX_OFED for Linux, WinOF-2 for Windows, and VMware. See the Driver Installation section.

Safety Precautions

The adapter is being installed in a system that operates with voltages that can be lethal. Before opening the case of the system, observe the following precautions to avoid injury and prevent damage to system components.

- Remove any metallic objects from your hands and wrists.
- Make sure to use only insulated tools.

- Verify that the system is powered off and is unplugged.
- It is strongly recommended to use an ESD strap or other antistatic devices.

Pre-Installation Checklist

- Unpack the ConnectX-6 Card; Unpack and remove the ConnectX-6 card. Check against the package contents list that all the parts have been sent. Check the parts for visible damage that may have occurred during shipping. Please note that the cards must be placed on an antistatic surface. For package contents please refer to Package Contents.

  > ⚠ **Warning**
  >
  > Please note that if the card is removed hastily from the antistatic bag, the plastic ziplock may harm the EMI fingers on the networking connector. Carefully remove the card from the antistatic bag to avoid damaging the EMI fingers.

- Shut down your system if active; Turn off the power to the system, and disconnect the power cord. Refer to the system documentation for instructions. Before you install the ConnectX-6 card, make sure that the system is disconnected from power.
- (Optional) Check the mounting bracket on the ConnectX-6 or PCIe Auxiliary Connection Card; If required for your system, replace the full-height mounting bracket that is shipped mounted on the card with the supplied low-profile bracket. Refer to Bracket Replacement Instructions.

Bracket Replacement Instructions

The ConnectX-6 card and PCIe Auxiliary Connection card are usually shipped with an assembled high-profile bracket. If this form factor is suitable for your requirements, you can skip the remainder of this section and move to Installation Instructions. If you need to replace the high-profile bracket with the short bracket that is included in the shipping box, please follow the instructions in this section.

> ⚠ **Warning**
>
> Due to risk of damaging the EMI gasket, it is not recommended to replace the bracket more than three times.

To replace the bracket you will need the following parts:

- The new brackets of the proper height
- The 2 screws saved from the removal of the bracket

Removing the Existing Bracket

1. Using a torque driver, remove the two screws holding the bracket in place.
2. Separate the bracket from the ConnectX-6 card.

> ⚠ **Warning**
>
> Be careful not to put stress on the LEDs on the adapter card.

3. Save the two screws.

Installing the New Bracket

1. Place the bracket onto the card until the screw holes line up.

> ⚠ **Warning**
>
> Do not force the bracket onto the adapter card.

2. Screw on the bracket using the screws saved from the bracket removal procedure above.

> ⚠ **Warning**
>
> Use a torque driver to apply up to 2 lbs-in torque on the screws.

Installation Instructions

This section provides detailed instructions on how to install your adapter card in a system.

Choose the installation instructions according to the ConnectX-6 configuration you have purchased.

OPNs	Installation Instructions
MCX651105A-EDAT MCX653105A-HDATMCX653106A-HDATMCX653105A-ECATMCX653106A-ECATMCX653105A-EFATMCX653106A-EFAT MCX683105AN-HDAT	**PCIe x8/16 Cards Installation Instructions**
MCX654105A-HCAT MCX654106A-HCATMCX654106A-ECAT	**Socket Direct (2x PCIe x16) Cards Installation Instructions**
MCX653105A-HDAL MCX653106A-HDAL	**Cards for Intel Liquid-Cooled Platforms Installation Instructions**

Cables and Modules

Cable Installation

1. All cables can be inserted or removed with the unit powered on.

2. To insert a cable, press the connector into the port receptacle until the connector is firmly seated.

1. Support the weight of the cable before connecting the cable to the adapter card. Do this by using a cable holder or tying the cable to the rack.

2. Determine the correct orientation of the connector to the card before inserting the connector. Do not try and insert the connector upside down. This may damage the adapter card.

3. Insert the connector into the adapter card. Be careful to insert the connector straight into the cage. Do not apply any torque, up or down, to the connector cage in the adapter card.

4. Make sure that the connector locks in place.

> **Note**
>
> When installing cables make sure that the latches engage.

> **Warning**
>
> Always install and remove cables by pushing or pulling the cable and connector in a straight line with the card.

3. After inserting a cable into a port, the Green LED indicator will light when the physical connection is established (that is, when the unit is powered on and a cable is plugged into the port with the other end of the connector plugged into a functioning port). See LED Interface under the Interfaces section.

4. After plugging in a cable, lock the connector using the latching mechanism particular to the cable vendor. When data is being transferred the Green LED will blink. See LED Interface under the Interfaces section.

5. Care should be taken as not to impede the air exhaust flow through the ventilation holes. Use cable lengths which allow for routing horizontally around to the side of the chassis before bending upward or downward in the rack.

6. To remove a cable, disengage the locks and slowly pull the connector away from the port receptacle. LED indicator will turn off when the cable is unseated.

Identifying the Card in Your System

On Linux

Get the device location on the PCI bus by running lspci and locating lines with the string “Mellanox Technologies”:

ConnectX-6 Card Configuration	lspci Command Output Example

Single-port Socket Direct Card (2x PCIe x16)	`[root@mftqa-009 ~]# lspci |grep mellanox -i`
	`a3:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]`
	`e3:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]`

Dual-port Socket Direct Card (2x PCIe x16)	``` [root@mftqa-009 ~]# lspci |grep mellanox -i 05:00.0 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6] 05:00.1 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6] 82:00.0 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6] 82:00.1 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6] ``` In the output example above, the first two rows indicate that one card is installed in a PCI slot with PCI Bus address 05 (hexadecimal), PCI Device number 00 and PCI Function number 0 and 1. The other card is installed in a PCI slot with PCI Bus address 82 (hexa-decimal), PCI Device number 00 and PCI Function number 0 and 1.Since the two PCIe cards are installed in two PCIe slots, each card gets a unique PCI Bus and Device number. Each of the PCIe x16 busses sees two network ports; in effect, the two physical ports of the ConnectX-6 Socket Direct adapter are viewed as four net devices by the system.
Single-port PCIe x8/x16 Card	``` [root@mftqa-009 ~]# lspci |grep mellanox -i 3:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6] ```
Dual-port PCIe x16 Card	``` [root@mftqa-009 ~]# lspci |grep mellanox -i 86:00.0 Network controller: Mellanox Technologies MT28908A0 Family [ConnectX-6] 86:00.1 Network controller: Mellanox Technologies MT28908A0 Family [ConnectX-6] ```

On Windows

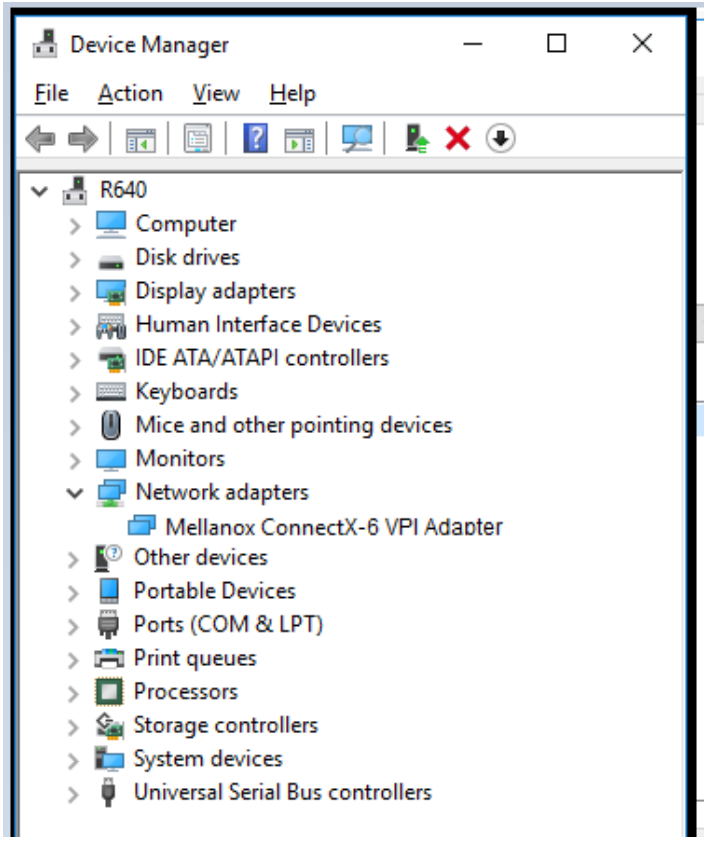
1. Open Device Manager on the server. Click **Start** => **Run**, and then enter **devmgmt.msc**.

2. Expand **System Devices** and locate your NVIDIA ConnectX-6 adapter card.

3. Right click the mouse on your adapter's row and select **Properties** to display the adapter card properties window.

4. Click the **Details** tab and select **Hardware Ids** (Windows 2012/R2/2016) from the **Property** pull-down menu.

   PCI Device (Example)



5. In the **Value** display box, check the fields VEN and DEV (fields are separated by '&'). In the display example above, notice the sub-string "PCI\VEN_15B3&DEV_1003": VEN is equal to 0x15B3 – this is the Vendor ID of NVIDIA; and DEV is equal to 1018 (for ConnectX-6) – this is a valid NVIDIA PCI Device ID.

Note

If the PCI device does not have a NVIDIA adapter ID, return to Step 2 to check another device.

Note

The list of NVIDIA PCI Device IDs can be found in the PCI ID repository at http://pci-ids.ucw.cz/read/PC/15b3.

PCIe x8/16 Cards Installation Instructions

Installing the Card

Note

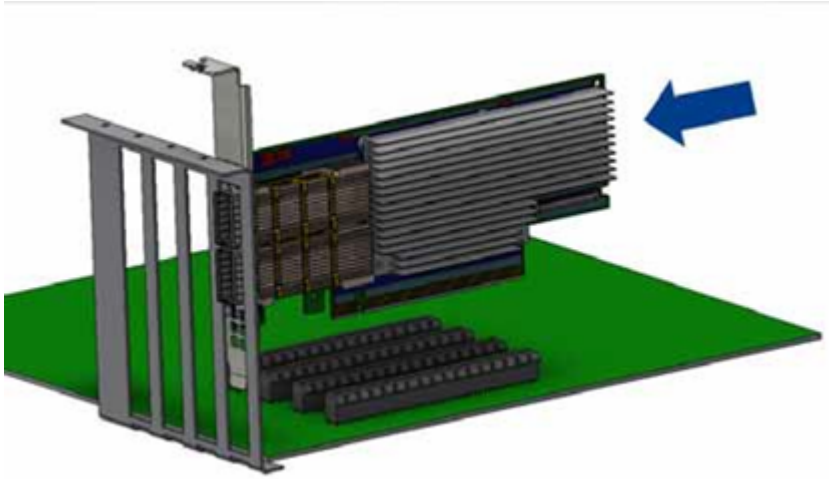
Applies to OPNs MCX651105A-EDAT, MCX654105A-HCAT, MCX654106A-HCAT, MCX683105AN-HDAT, MCX653106A-ECAT and MCX653105A-ECAT .

Note

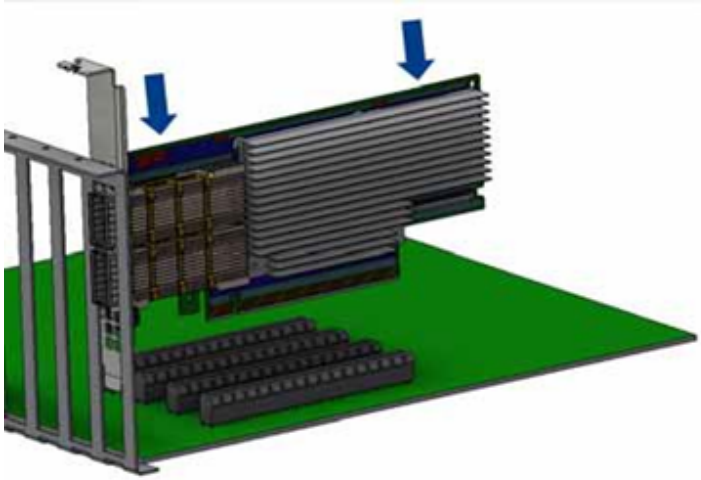
Please make sure to install the ConnectX-6 cards in a PCIe slot that is capable of supplying the required power and airflow as stated in Specifications.

➢ Connect the adapter Card in an available PCI Express x16 slot in the chassis.

Step 1: Locate an available PCI Express x16 slot and insert the adapter card to the chassis.



Step 2: Hold the card by its edges and bracket, avoiding contact with the component or print side. Apply even pressure to both edges and carefully insert the adapter into the PCI Express slot until firmly seated.

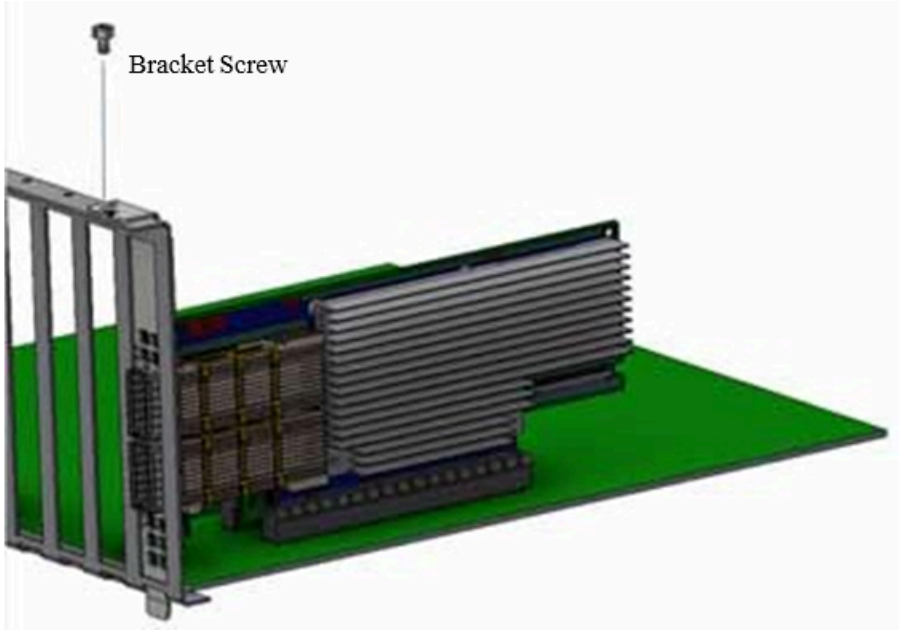


⚠ Warning

Avoid touching the components on the adapter card. Avoid applying excessive force when seating the card, as this could damage the system or the adapter card components.

Secure the adapter card to the chassis.

Step 1: Secure the bracket to the chassis with the bracket screw.



Uninstalling the Card

Safety Precautions

The adapter is installed in a system that operates with voltages that can be lethal. Before uninstalling the adapter card, please observe the following precautions to avoid injury and prevent damage to system components.

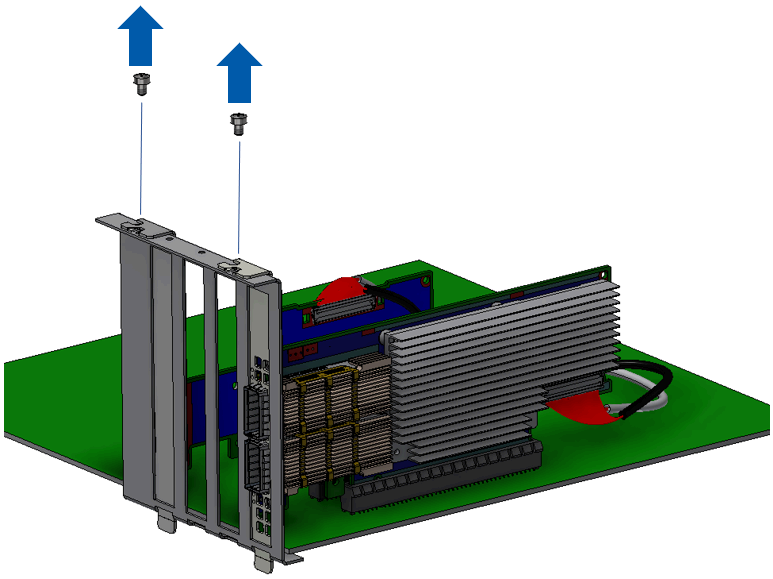
1. Remove any metallic objects from your hands and wrists.
2. It is strongly recommended to use an ESD strap or other antistatic devices.
3. Turn off the system and disconnect the power cord from the server.

Card Removal

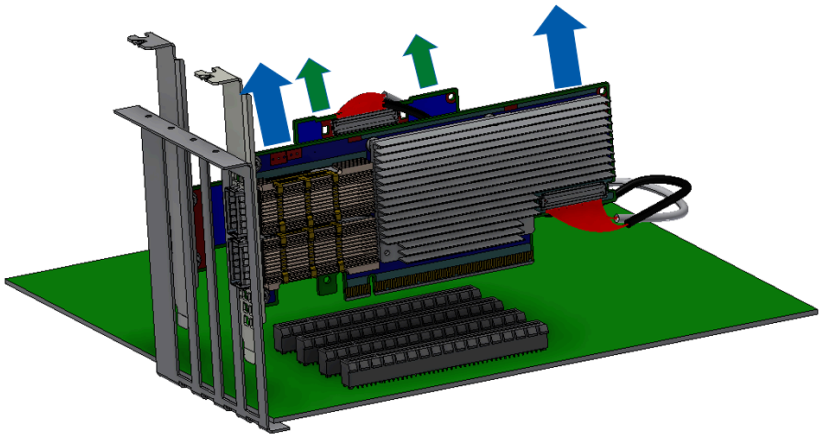
Info

Please note that the following images are for illustration purposes only.

1. Verify that the system is powered off and unplugged.
2. Wait 30 seconds.
3. To remove the card, disengage the retention mechanisms on the bracket (clips or screws).



4. Holding the adapter card from its center, gently pull the ConnectX-6 and Auxiliary Connections cards out of the PCI Express slot.



Socket Direct (2x PCIe x16) Cards Installation Instructions

The hardware installation section uses the terminology of white and black harnesses to differentiate between the two supplied cables. Due to supply chain variations, some cards may be supplied with two black harnesses instead. To clarify the difference between these two harnesses, one black harness was marked with a “WHITE” label and the other with a “BLACK” label.

The Cabline harness marked with "WHITE" label should be connected to the connector on the ConnectX-6 and PCIe card engraved with “White Cable” while the one marked with "BLACK" label should be connected to the connector on the ConnectX-6 and PCIe card engraved with “Black Cable”.

Note

The harnesses' minimal bending radius is 10[mm].

Installing the Card

Note

Applies to MCX654105A-HCAT, MCX654106A-HCAT and MCX654106A-ECAT.

Note

The installation instructions include steps that involve a retention clip to be used while connecting the Cabline harnesses to the cards. Please note that this is an optional accessory.

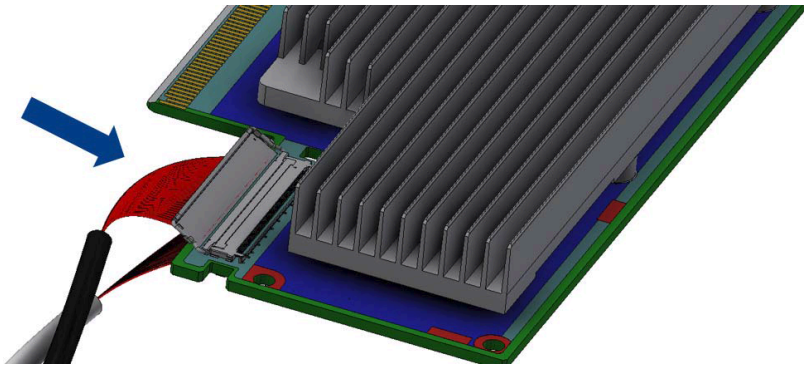
Note

Please make sure to install the ConnectX-6 cards in a PCIe slot that is capable of supplying the required power and airflow as stated in Specifications.

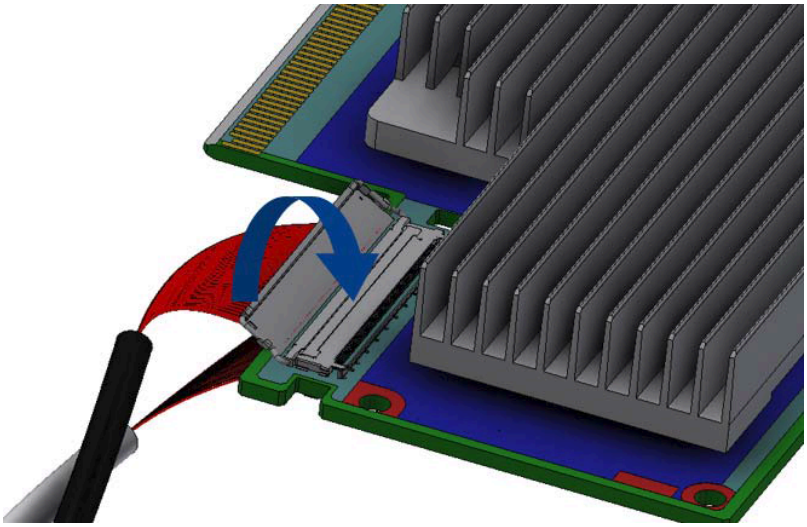
Step 1: Slide the black and white Cabline CA-II Plus harnesses through the retention clip while making sure the clip opening is facing the plugs.



Step 2: Plug the Cabline CA-II Plus harnesses on the ConnectX-6 adapter card while paying attention to the color-coding. As indicated on both sides of the card; plug the black harness to the component side and the white harness to the print side.



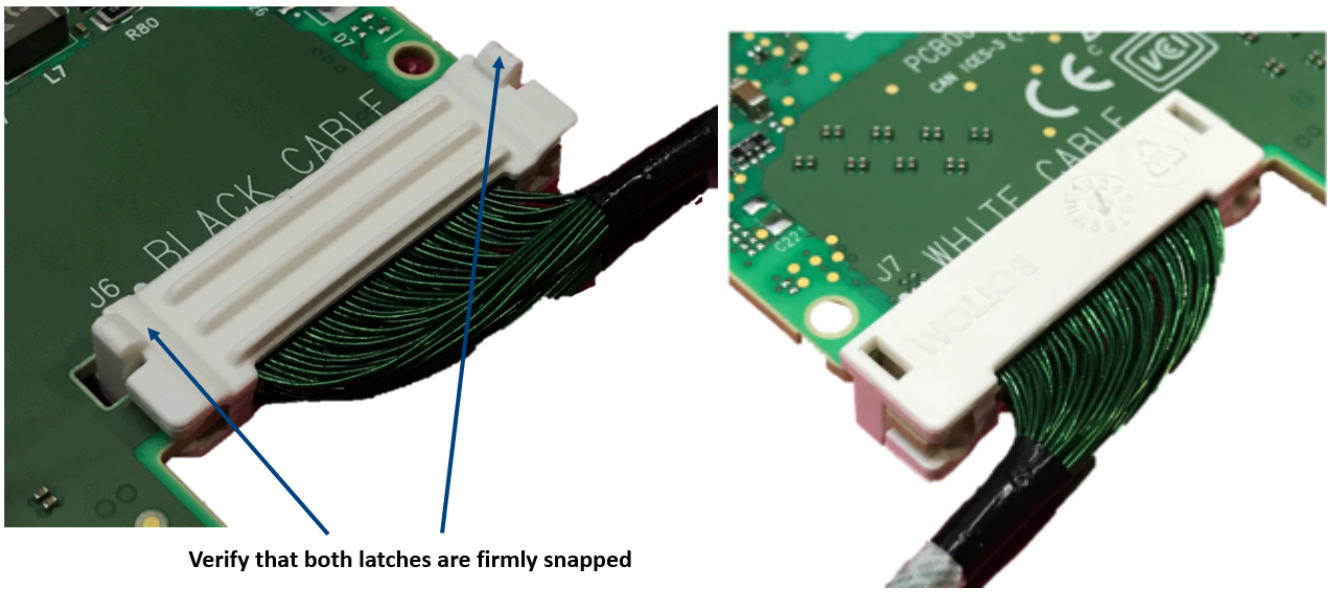
Step 2: Verify the plugs are locked.



Step 3: Slide the retention clip latches through the cutouts on the PCB. The latches should face the annotation on the PCB.



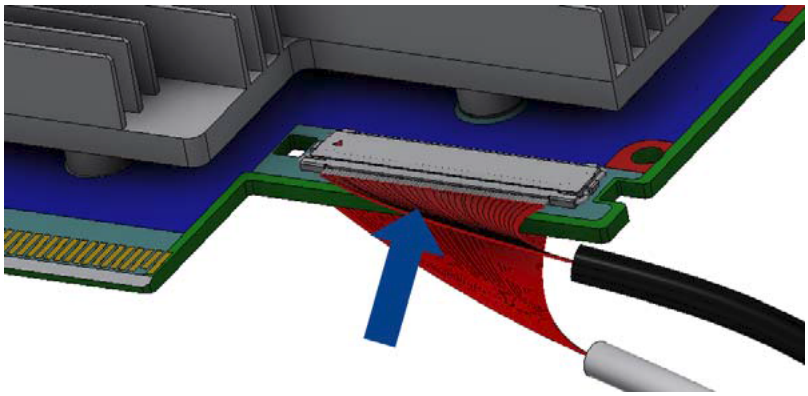
Step 4: Clamp the retention clip. Verify both latches are firmly locked.



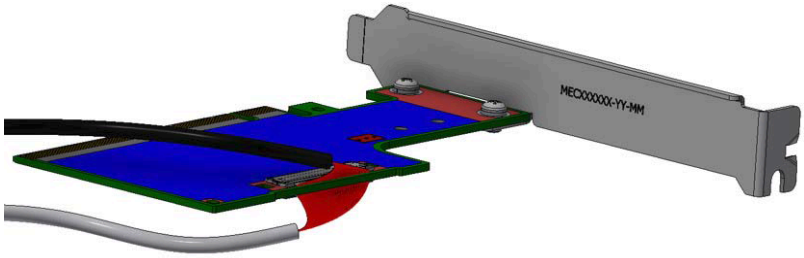
Step 5: Slide the Cabline CA-II Plus harnesses through the retention clip. Make sure that the clip opening is facing the plugs.



Step 6: Plug the Cabline CA-II Plus harnesses on the PCIe Auxiliary Card. As indicated on both sides of the Auxiliary connection card; plug the black harness to the component side and the white harness to the print side.



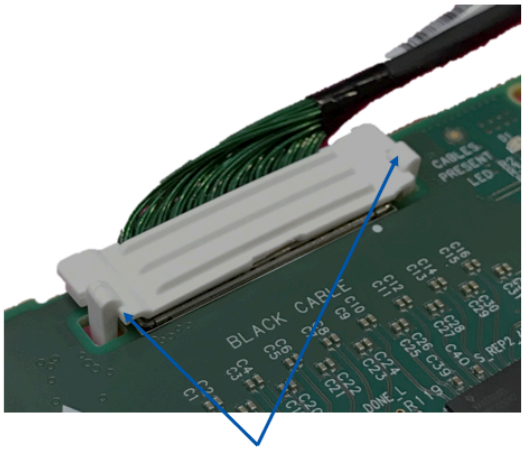
Step 7: Verify the plugs are locked.



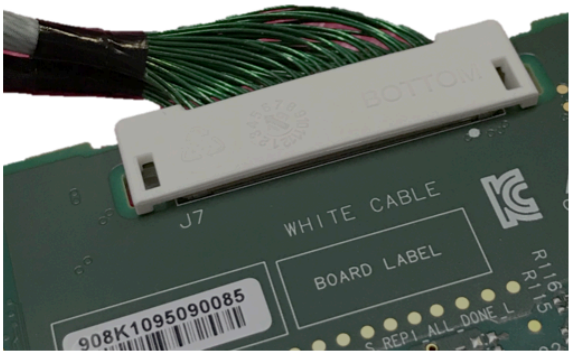
Step 8: Slide the retention clip through the cutouts on the PCB. Make sure latches are facing "Black Cable" annotation as seen in the below picture.



Step 9: Clamp the retention clip. Verify both latches are firmly locked.

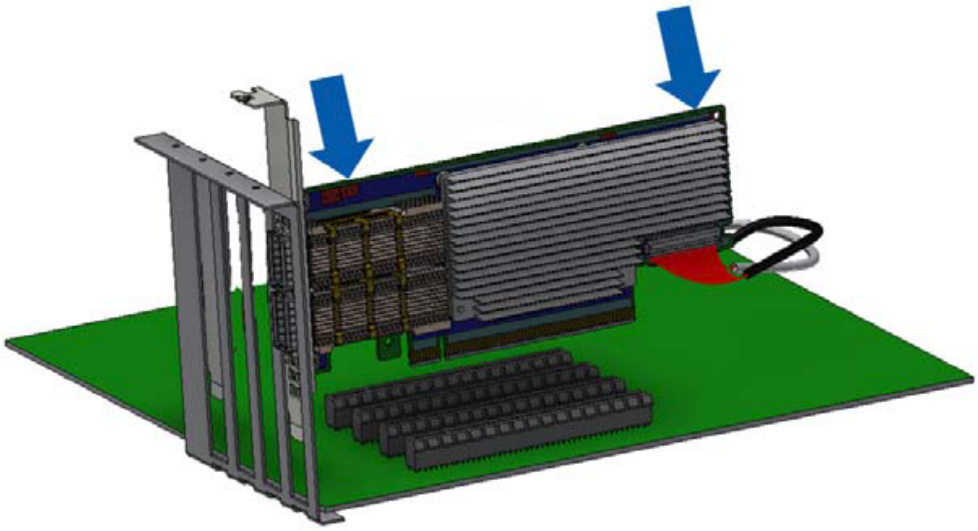


Verify that both latches are firmly snapped



Step 1: Locate two available PCI Express x16 slots.

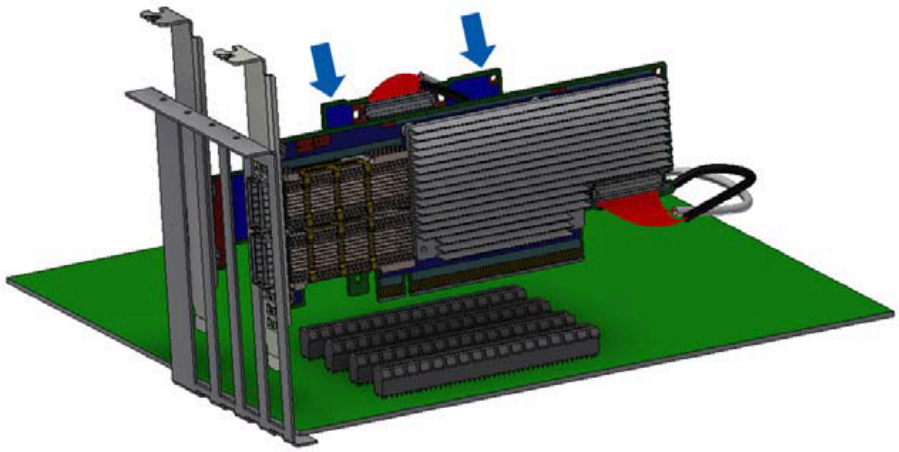
Step 2: Hold the card by its edges and bracket, avoiding contact with the component or print side. Apply even pressure to both edges and carefully insert the adapter into the PCI Express slot until firmly seated.



Warning

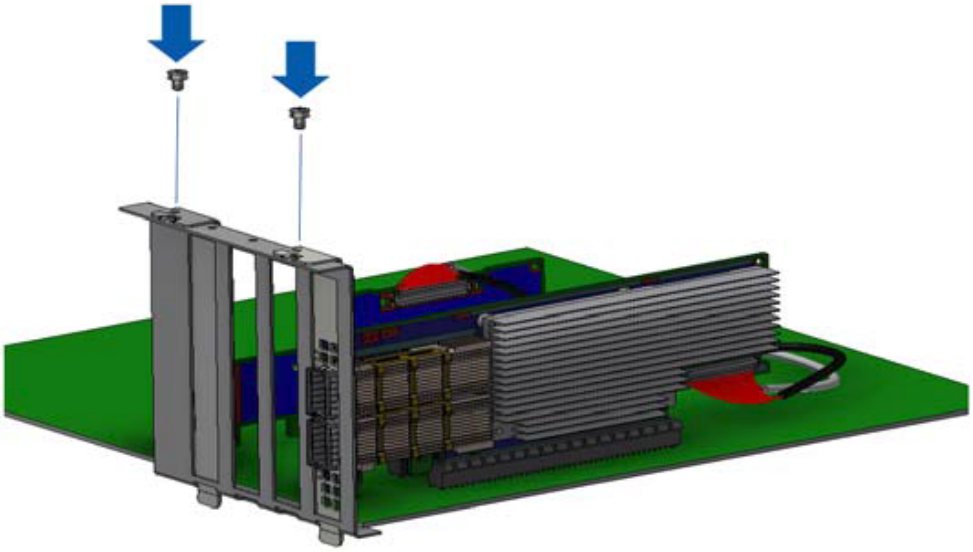
- Avoid touching the components on the adapter card.
- Avoid applying excessive force when seating the card, as this could damage the system or the adapter card components..

Step 3: Applying even pressure at both corners of the cards, insert the Auxiliary Connection card in the PCI Express slots until firmly seated.



.

Step 1: Secure the brackets to the chassis with the bracket screw.



Uninstalling the Card

Safety Precautions

The adapter is installed in a system that operates with voltages that can be lethal. Before uninstalling the adapter card, please observe the following precautions to avoid injury and prevent damage to system components.

1. Remove any metallic objects from your hands and wrists.
2. It is strongly recommended to use an ESD strap or other antistatic devices.
3. Turn off the system and disconnect the power cord from the server.

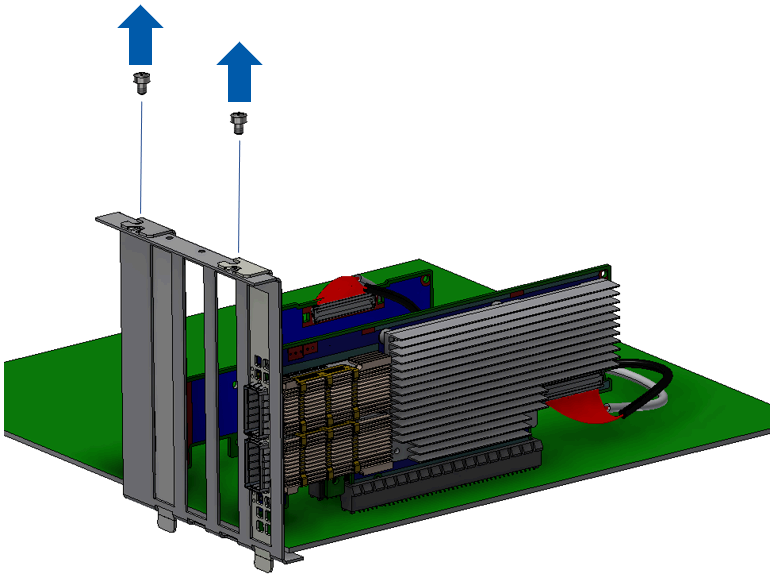
Card Removal

> **Info**
>
> Please note that the following images are for illustration purposes only.

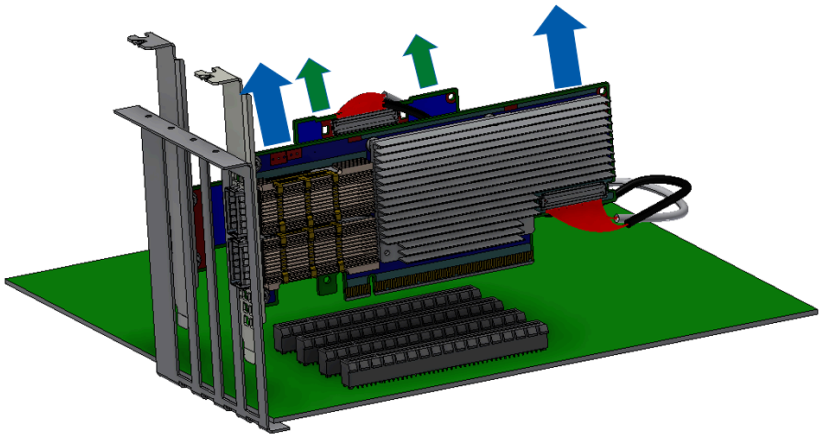
1. Verify that the system is powered off and unplugged.

2. Wait 30 seconds.

3. To remove the card, disengage the retention mechanisms on the bracket (clips or screws).



4. Holding the adapter card from its center, gently pull the ConnectX-6 and Auxiliary Connections cards out of the PCI Express slot.



Cards for Intel Liquid-Cooled Platforms Installation Instructions

The below instructions apply to ConnectX-6 cards designed for Intel liquid-cooled platforms with ASIC interposer cooling mechanism. **OPNs: MCX653105A-HDAL and**

MCX653106A-HDAL.

> **Note**
>
> The below figures are for illustration purposes only.

> **Warning**
>
> The below instructions should be used in conjunction with the server's documentation.

Installing the Card

> **Note**
>
> Please make sure the system is capable of supplying the required power as stated in Specifications.

> **Warning**
>
> Pay extra attention to the black bumpers located on the print side of the card. Failure to do so may harm the bumpers.

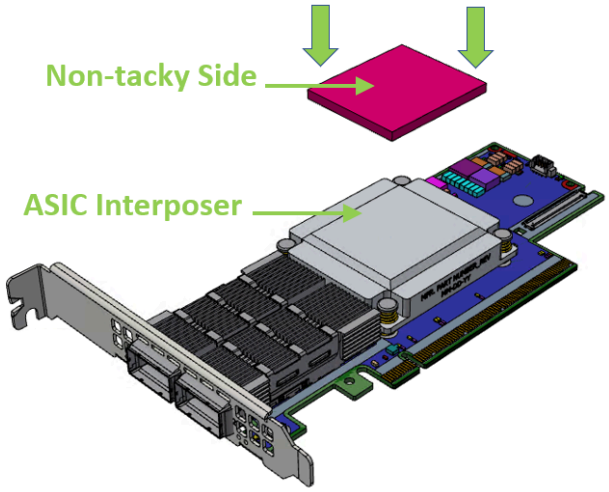
➢ **Apply the supplied thermal pad (one of the two) on top of the ASIC interposer or onto the coldplate.**

⚠ Warning

The thermal pads are shipped with two protective liners covering the pad on both sides. It is highly important to peel the liners as instructed below prior to applying them to the card.

1. Gently peel the liner from the thermal pad's tacky side.
2. Carefully apply the thermal pad on the cool block (ASIC interposer) while ensuring it thoroughly covers it. **Extra care should be taken not to damage the pad.**

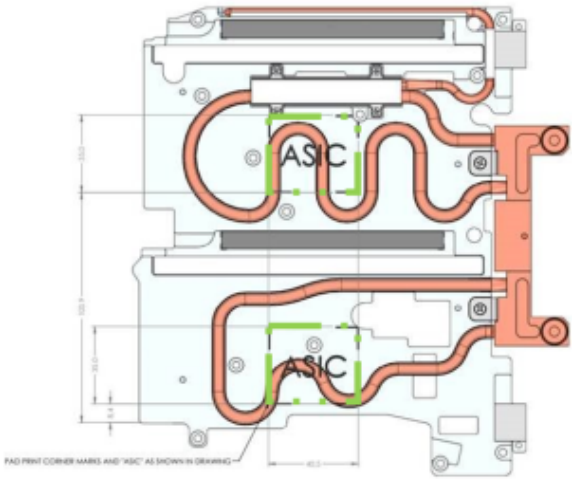
   The thermal pad should be applied on the cool block from its **tacky** (wet) side. The pad should be applied with its non-tacky side facing up.



OR

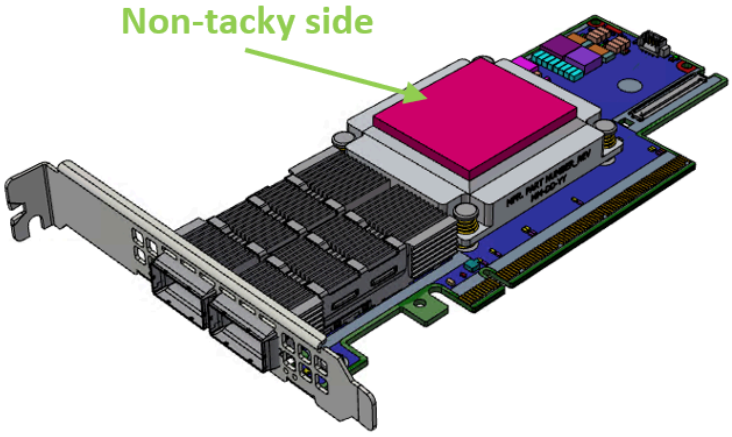
Carefully apply the thermal pad on the coldplate while ensuring it thoroughly covers it. The below figure indicates the position of the thermal pad. Extra care should be taken not to damage the pad.

The thermal pad should be applied on the coldplate from its **tacky** (wet) side. The pad should be applied with its non-tacky side facing up.

1. 

2. Ensure the thermal pad is in place and intact.

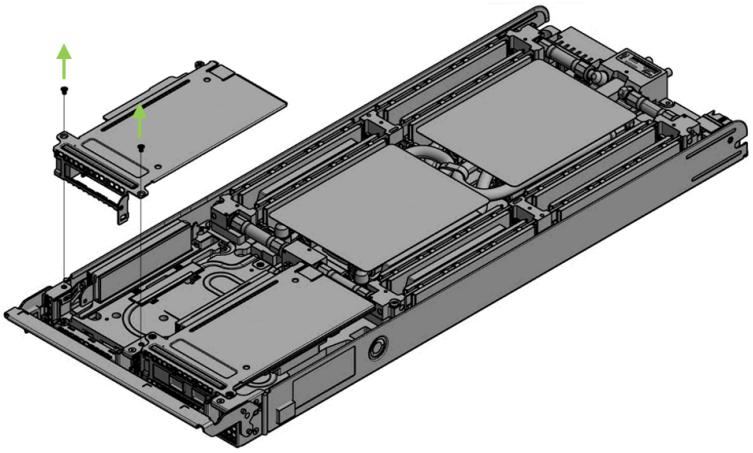
3. Once the thermal pad is applied to the ASIC interposer, the non-tacky side should be visible on the card's faceplate.



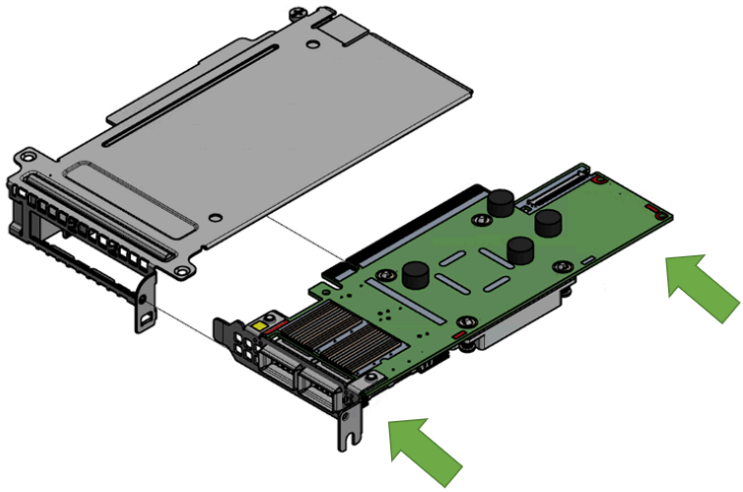
4. Gently peel the liner of the pad's non-tacky side visible on the card's faceplate. Failure to do so may degrade the thermal performance of the product.

➢ **Install the adapter into the riser and attach the card to the PCIe x16 slot.**

1. Disengage the adapter riser from the blade. Please refer to the blade documentation for instructions.

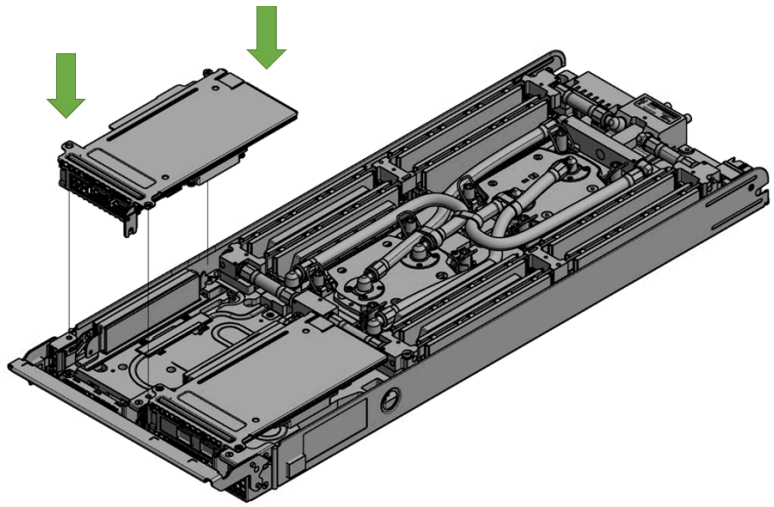


2. Applying even pressure at both corners of the card, insert the adapter card into the adapter riser until firmly seated. Care must be taken to not harm the black bumpers located on the print side of the card.

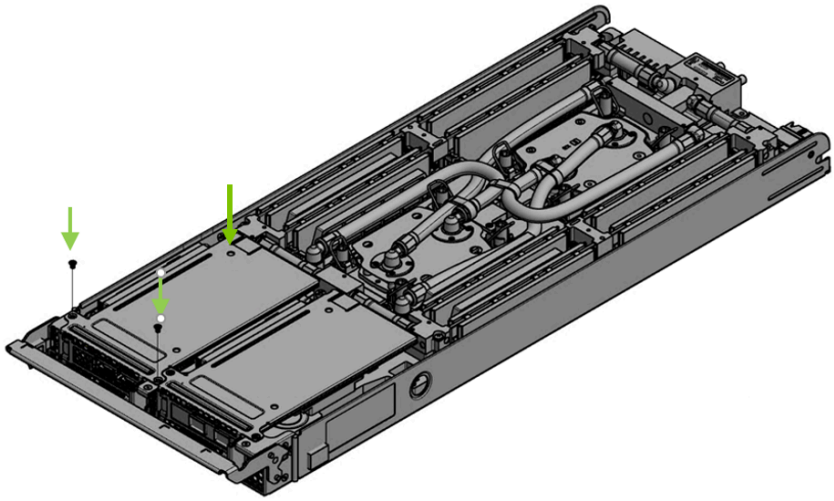


Vertically insert the riser that populates the adapter card into the server blade.

1. Applying even pressure on the riser, gently insert the riser into the server.



2. Secure the riser with the supplied screws. Please refer to the server blade documentation for more information.



Driver Installation

Please use the relevant driver installation section.

> **Note**
>
> ConnectX-6 Socket Direct cards 2x PCIe x16 (OPNs: MCX654106A-HCAT and MCX654106A-ECAT) are not supported in Windows and WinOF-2.

- Linux Driver Installation
- Windows Driver Installation
- VMware Driver Installation

Linux Driver Installation

This section describes how to install and test the MLNX_OFED for Linux package on a single server with a NVIDIA ConnectX-6 adapter card installed.

Prerequisites

Requirements	Description
Platforms	A server platform with a ConnectX-6 InfiniBand/Ethernet adapter card installed.
Required Disk Space for Installation	1GB
Operating System	Linux operating system. For the list of supported operating system distributions and kernels, please refer to the *MLNX_OFED Release Notes*

Requirements	Description
Installer Privileges	The installation requires administrator (root) privileges on the target machine.

Downloading NVIDIA OFED

1. Verify that the system has a NVIDIA network adapter installed by running lscpi command. The below table provides output examples per ConnectX-6 card configuration.

ConnectX-6 Card Configuration	

Single-port Socket Direct Card (2x PCIe x16)	`[root@mftqa-009 ~]# lspci |grep mellanox -i`
	`a3:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]`
	`e3:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]`

ConnectX-6 Card Configuration	

Dual-port Socket Direct Card (2x PCIe x16)	`[root@mftqa-009 ~]# lspci |grep mellanox -i`
	`05:00.0 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6]`
	`05:00.1 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6]`
	`82:00.0 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6]`
	`82:00.1 Infiniband controller: Mellanox Technologies MT28908A0 Family [ConnectX-6]`
	In the output example above, the first two rows indicate that one card is installed in a PCI slot with PCI Bus address 05 (hexadecimal), PCI Device number 00 and PCI Function number 0 and 1. The other card is installed in a PCI slot with PCI Bus address 82 (hexadecimal), PCI Device number 00 and PCI Function number 0 and 1.Since the two PCIe cards are installed in two PCIe slots, each card gets a unique PCI Bus and Device number. Each of the PCIe x16 busses sees two network ports; in effect, the two physical ports of the ConnectX-6 Socket Direct adapter are viewed as four net devices by the system.

Single-port PCIe x16 Card	`[root@mftqa-009 ~]# lspci |grep mellanox -ia` `3:00.0 Infiniband controller: Mellanox Technologies MT28908 Family [ConnectX-6]`

ConnectX-6 Card Configuration	

Dual-port PCIe x16 Card	`[root@mftqa-009 ~]# lspci |grep mellanox -ia`
	`86:00.0 Network controller: Mellanox Technologies MT28908A0 Family [ConnectX-6]`
	`86:00.1 Network controller: Mellanox Technologies MT28908A0 Family [ConnectX-6]`

For Linux driver installation, please refer to NVIDIA DOCA Installation Guide for Linux.

Windows Driver Installation

> **Note**
>
> Windows driver is currently not supported in the following ConnectX-6 OPNs:
>
> - MCX654106A-HCAT
> - MCX654106A-ECAT

For Windows, download and install the latest WinOF-2 for Windows software package available via the NVIDIA website at: WinOF-2 webpage. Follow the installation instructions included in the download package (also available from the download page).

The snapshots in the following sections are presented for illustration purposes only. The installation interface may slightly vary, depending on the operating system in use.

Software Requirements

For the full list of supported operating systems, please refer to the WinOF-2 Documentation -> Release Notes.

Downloading WinOF-2 Driver

To download the .exe file according to your Operating System, please follow the steps below:

1. Obtain the machine architecture.

    1. To go to the Start menu, position your mouse in the bottom-right corner of the Remote Desktop of your screen.

    2. Open a CMD console (Click Task Manager-->File --> Run new task and enter CMD).

    3. Enter the following command.

        ```
        echo %PROCESSOR_ARCHITECTURE%
        ```

        Note

        On an x64 (64-bit) machine, the output will be “AMD64”.

2. Go to the WinOF-2 web page at: https://www.nvidia.com/en-us/networking/ > Products > Software > InfiniBand Drivers (Learn More) > Nvidia WinOF-2.

3. Download the .exe image according to the architecture of your machine (see Step 1).

    The name of the .exe is in the following format: MLNX_WinOF2-<version>_<arch>.exe.

Note

Installing the incorrect .exe file is prohibited. If you do so, an error message will be displayed.

For example, if you install a 64-bit .exe on a 32-bit machine, the wizard will display the following (or a similar) error message: “The installation package is not supported by this processor type. Contact your vendor”

Installing WinOF-2 Driver

The snapshots in the following sections are for illustration purposes only. The installation interface may slightly vary, depending on the used operating system.

This section provides instructions for two types of installation procedures, and both require administrator privileges:

- Attended Installation

  An installation procedure that requires frequent user intervention.

- Unattended Installation

  An automated installation procedure that requires no user intervention.

Attended Installation

The following is an example of an installation session.

1. Double click the .exe and follow the GUI instructions to install MLNX_WinOF2.
2. **[Optional]** Manually configure your setup to contain the logs option (replace “LogFile” with the relevant directory).

```
MLNX_WinOF2_<revision_version>_All_Arch.exe /v"/l*vx
[LogFile]"
```

3. **[Optional]** If you do not want to upgrade your firmware version (i.e., MT_SKIPFWUPGRD default value is False).

```
MLNX_WinOF2_<revision_version>_All_Arch.exe /v" MT_SKIPFWUPGRD=1"
```

4. **[Optional]** If you do not want to install the Rshim driver, run.

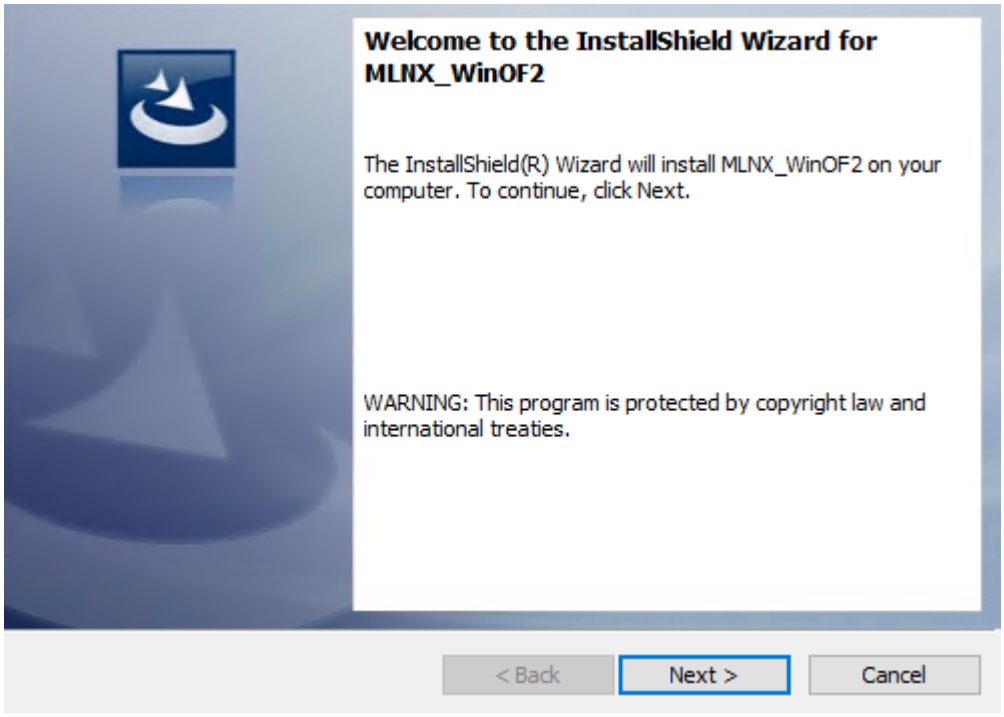
```
MLNX_WinOF2_<revision_version>_All_Arch.exe /v"
MT_DISABLE_RSHIM_INSTALL=1"
```

> **Note**
>
> The Rshim driver installanion will fail if a prior Rshim driver is already installed. The following fail message will be displayed in the log:
>
> "ERROR!!! Installation failed due to following errors: MlxRshim drivers installation disabled and MlxRshim drivers Installed, Please remove the following oem inf files from driver store: <oem inf list>"

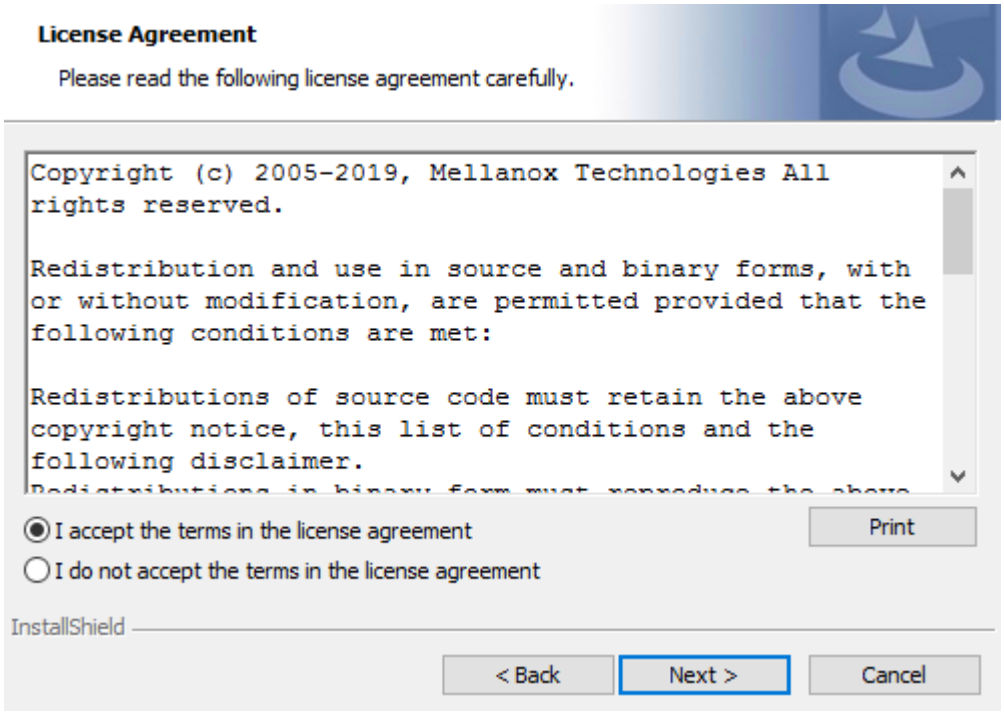
5. **[Optional]** If you want to skip the check for unsupported devices, run.

```
MLNX_WinOF2_<revision_version>_All_Arch.exe /v"
SKIPUNSUPPORTEDDEVCHECK=1"
```

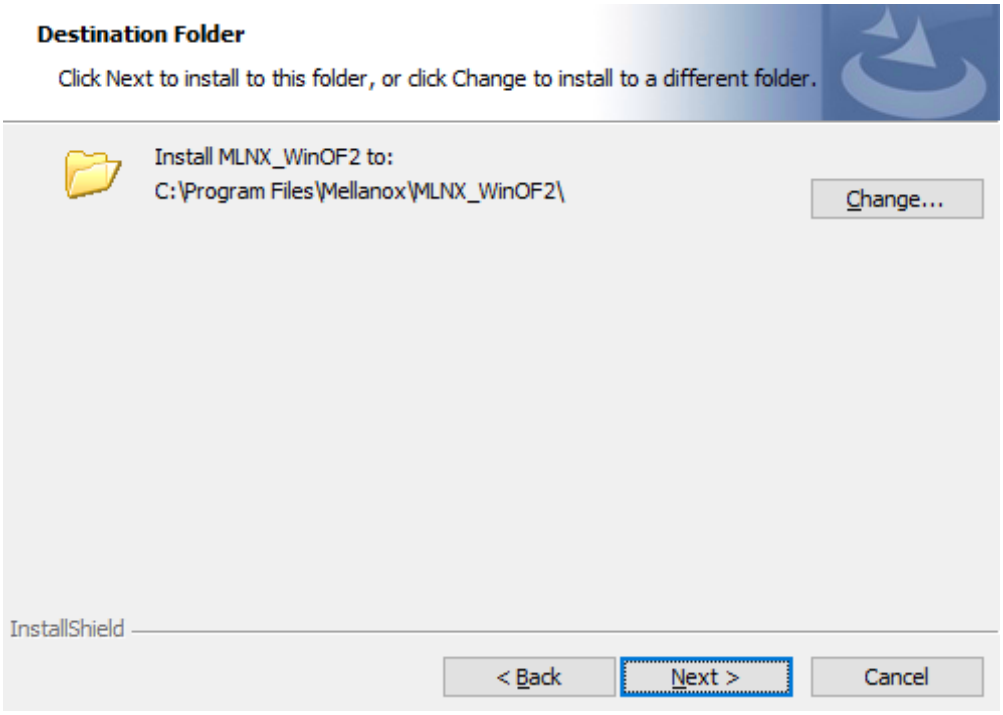
6. Click Next in the Welcome screen.



7. Read and accept the license agreement and click Next.

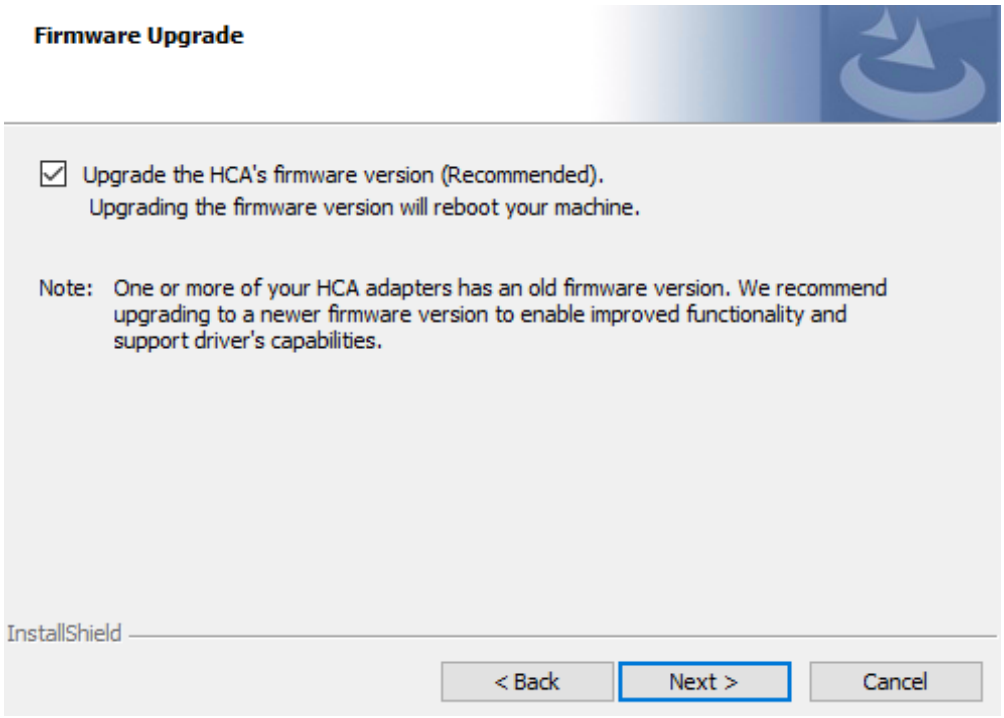


8. Select the target folder for the installation.

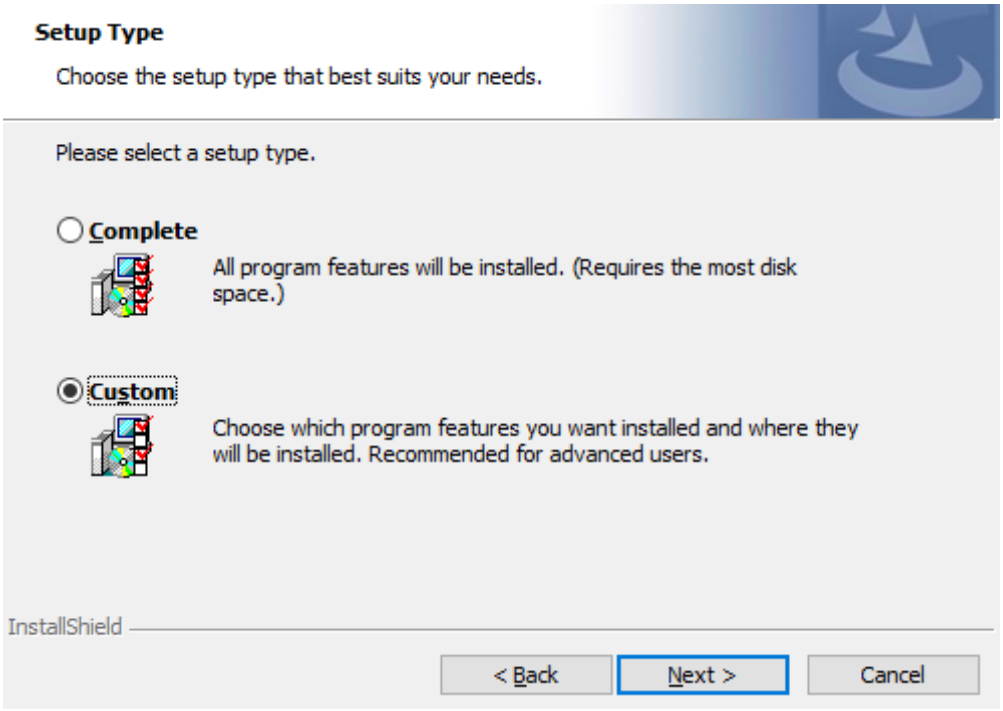


9. The firmware upgrade screen will be displayed in the following cases:

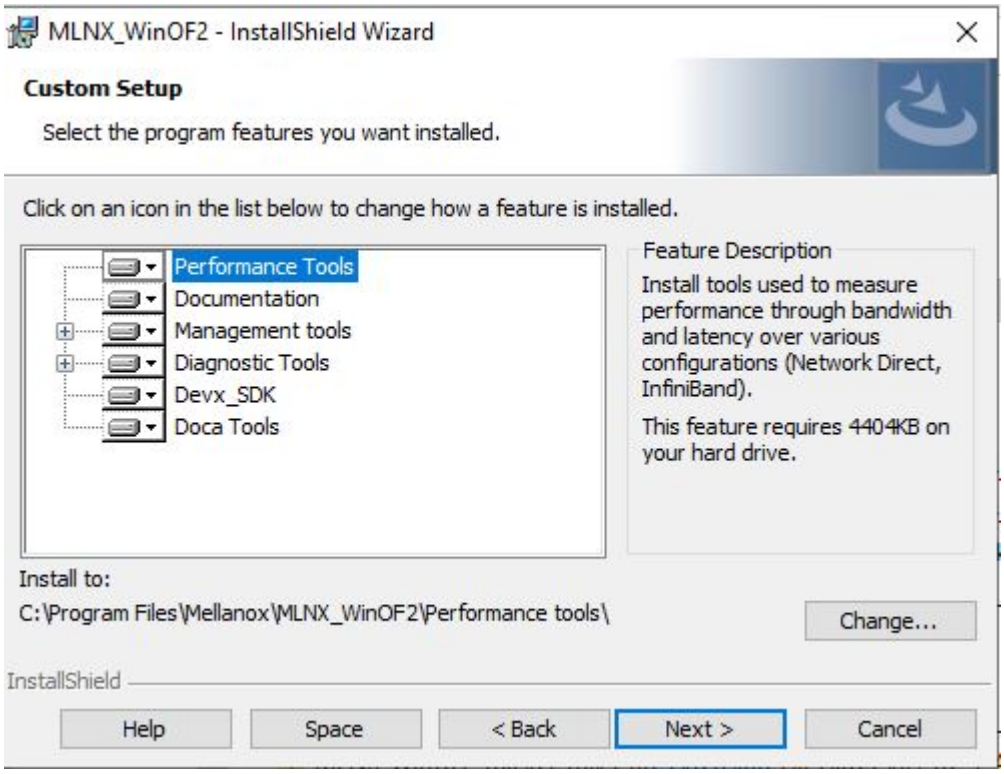
    - If the user has an OEM card. In this case, the firmware will not be displayed.
    - If the user has a standard NVIDIA® card with an older firmware version, the firmware will be updated accordingly. However, if the user has both an OEM card and a NVIDIA® card, only the NVIDIA® card will be updated.



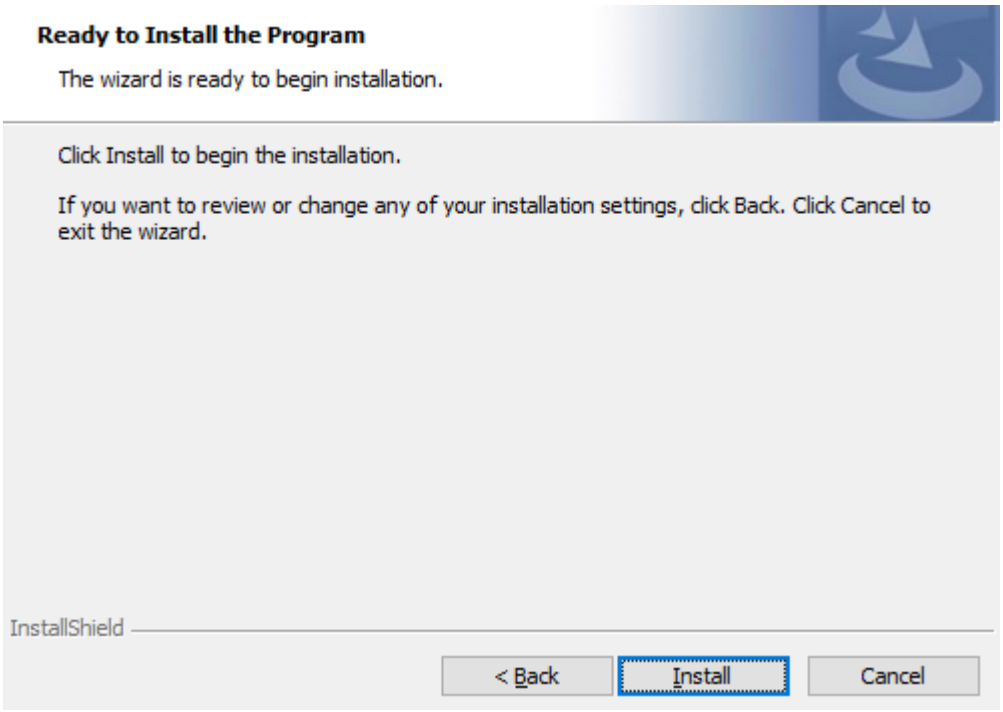
10. Select a Complete or Custom installation, follow Step a onward.



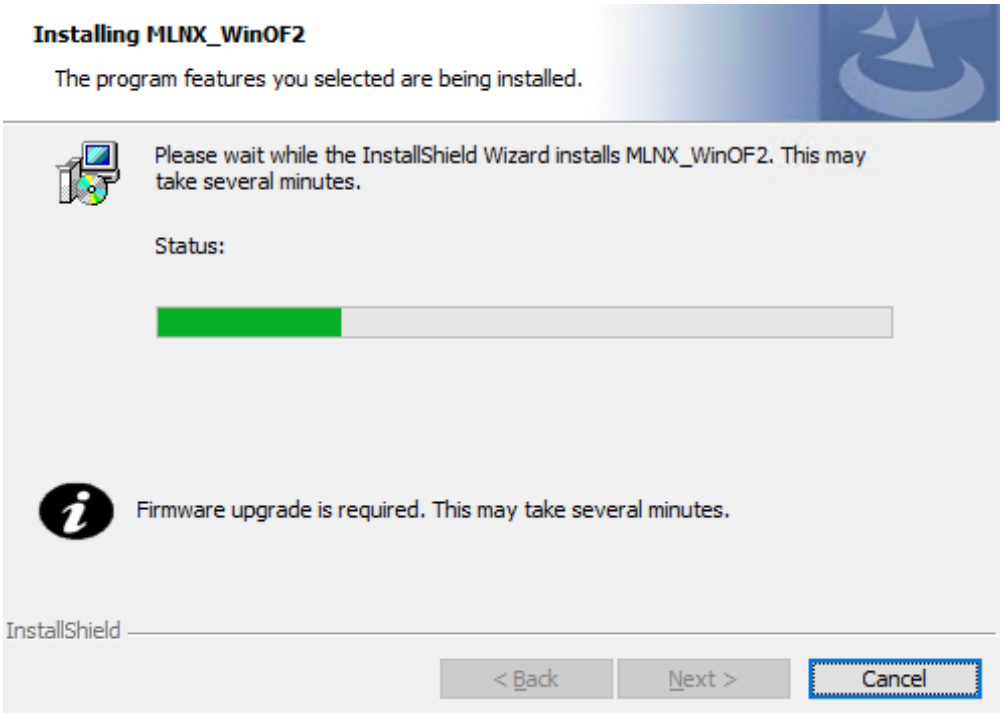
1. Select the desired feature to install:
    - Performances tools - install the performance tools that are used to measure performance in user environment
    - Documentation - contains the User Manual and Release Notes
    - Management tools - installation tools used for management, such as mlxstat
    - Diagnostic Tools - installation tools used for diagnostics, such as mlx5cmd
2. Click Next to install the desired tools.



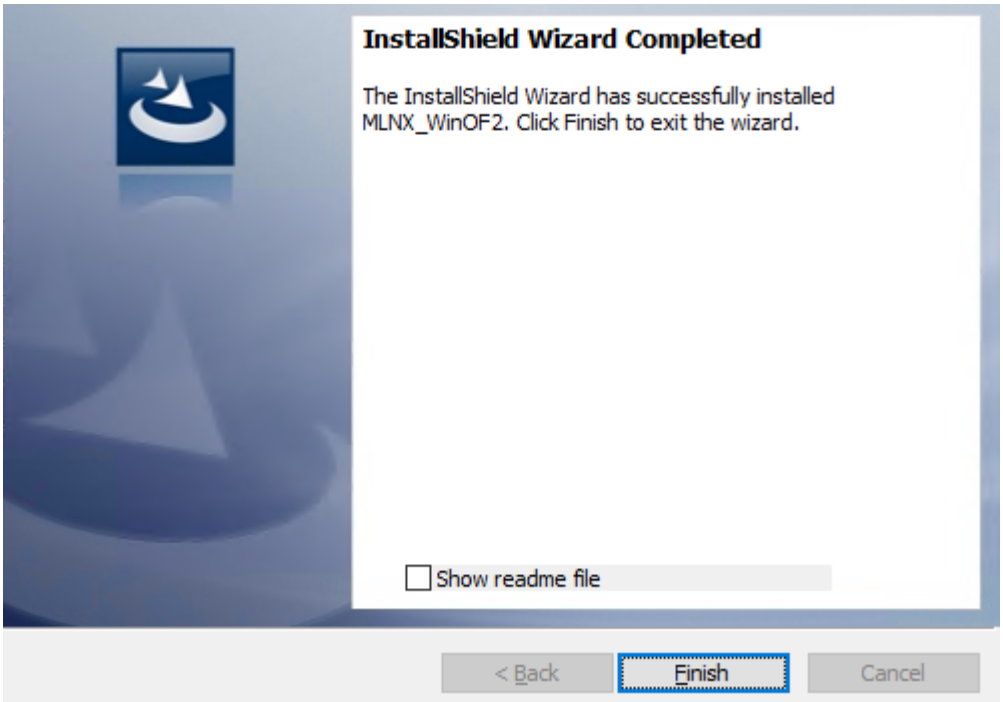
11. Click Install to start the installation.



12. In case firmware upgrade option was checked in Step 7, you will be notified if a firmware upgrade is required (see).



13. Click Finish to complete the installation.



Unattended Installation

Note

If no reboot options are specified, the installer restarts the computer whenever necessary without displaying any prompt or warning to the user.

To control the reboots, use the */norestart* or */forcerestart* standard command-line options.

The following is an example of an unattended installation session.

1. Open a CMD console**->** Click Start**->** Task Manager File**->** Run new task**->** and enter CMD.
2. Install the driver. Run:

```
MLNX_WinOF2-[Driver/Version]_<revision_version>_All_-Arch.exe
/S /v/qn
```

3. **[Optional]** Manually configure your setup to contain the logs option:

```
MLNX_WinOF2-[Driver/Version]_<revision_version>_All_-Arch.exe
/S /v/qn /v"/l*vx [LogFile]"
```

4. **[Optional]** if you wish to control whether to install ND provider or not (i.e., *MT_NDPROPERTY default value is True*).

```
MLNX_WinOF2-[Driver/Version]_<revision_version>_All_Arch.exe
/vMT_NDPROPERTY=1
```

5. **[Optional]** If you do not wish to upgrade your firmware version (i.e.,*MT_SKIPFWUPGRD default value is False*).

```
MLNX_WinOF2-[Driver/Version]_<revision_version>_All_Arch.exe
/vMT_SKIPFWUPGRD=1
```

6. **[Optional]** If you do not want to install the Rshim driver, run.

```
MLNX_WinOF2_<revision_version>_All_Arch.exe /v"
MT_DISABLE_RSHIM_INSTALL=1"
```

> **Note**
>
> The Rshim driver installanion will fail if a prior Rshim driver is already installed. The following fail message will be displayed in the log:
>
> "ERROR!!! Installation failed due to following errors: MlxRshim drivers installation disabled and MlxRshim drivers Installed, Please remove the following oem inf files from driver store: <oem inf list>"

7. **[Optional]** If you want to enable the default configuration for Rivermax, run.

```
MLNX_WinOF2_<revision_version>_All_Arch.exe /v"MT_RIVERMAX=1 /l*vx
C:\Users\<user>\log.txt "
```

8. **[Optional]** If you want to skip the check for unsupported devices, run/

```
MLNX_WinOF2_<revision_version>_All_Arch.exe /v"
SKIPUNSUPPORTEDDEVCHECK=1"
```

Firmware Upgrade

If the machine has a standard NVIDIA® card with an older firmware version, the firmware will be automatically updated as part of the NVIDIA® WinOF-2 package installation. For information on how to upgrade firmware manually, please refer to MFT User Manual.

If the machine has a DDA (pass through) facility, firmware update is supported only in the Host. Therefore, to update the firmware, the following must be performed:

1. Return the network adapters to the Host.
2. Update the firmware according to the steps in the MFT User Manual.
3. Attach the adapters back to VM with the DDA tools.

VMware Driver Installation

This section describes VMware Driver Installation.

Software Requirements

Requirement	Description
Platforms	A server platform with an adapter card based on ConnectX®-6 (InfiniBand/EN) (firmware: fw-ConnectX6)
Operating System	ESXi 6.5
Installer Privileges	The installation requires administrator privileges on the target machine.

Installing NATIVE ESXi Driver for VMware vSphere

Note

Please uninstall all previous driver packages prior to installing the new version.

To install the driver:

1. Log into the ESXi server with root permissions.
2. Install the driver.

```
#> esxcli software vib install –d <path>/<bundle_file>
```

Example:

```
#> esxcli software vib install -d /tmp/MLNX-NATIVE-ESX-
ConnectX-4-5_4.16.8.8-10EM-650.0.0.4240417.zipesxcli
```

3. Reboot the machine.
4. Verify the driver was installed successfully.

```
esxcli software vib list | grep nmlx
nmlx5-core      4.16.8.8-10EM.650.0.0.4240417    MEL
PartnerSupported 2017-01-31
nmlx5-rdma      4.16.8.8-10EM.650.0.0.4240417    MEL
PartnerSupported 2017-01-31
```

Note

After the installation process, all kernel modules are loaded automatically upon boot.

Removing Earlier NVIDIA Drivers

Note

Please unload the previously installed drivers before removing them.

To remove all the drivers:

1. Log into the ESXi server with root permissions.
2. List all the existing NATIVE ESXi driver modules. (See Step 4 in Installing NATIVE ESXi Driver for VMware vSphere.)
3. Remove each module:

```
#> esxcli software vib remove -n nmlx5-rdma
#> esxcli software vib remove -n nmlx5-core
```

Note

To remove the modules, you must run the command in the same order as shown in the example above.

4. Reboot the server.

Firmware Programming

1. Download the VMware bootable binary images v4.6.0 from the Firmware Tools (MFT) site.

    1. ESXi 6.5 File: mft-4.6.0.48-10EM-650.0.0.4598673.x86_64.vib

    2. MD5SUM: 0804cffe30913a7b4017445a0f0adbe1

2. Install the image according to the steps described in the *MFT User Manual*.

> **Note**
>
> The following procedure requires custom boot image downloading, mounting and booting from a USB device.

Troubleshooting

General Troubleshooting

Server unable to find the adapter	• Ensure that the adapter is placed correctly • Make sure the adapter slot and the adapter are compatible Install the adapter in a different PCI Express slot • Use the drivers that came with the adapter or download the latest • Make sure your motherboard has the latest BIOS • Try to reboot the server
The adapter no longer works	• Reseat the adapter in its slot or a different slot, if necessary • Try using another cable • Reinstall the drivers for the network driver files may be damaged or deleted • Reboot the server
Adapters stopped working after installing another adapter	• Try removing and re-installing all adapters • Check that cables are connected properly • Make sure your motherboard has the latest BIOS
Link indicator light is off	• Try another port on the switch • Make sure the cable is securely attached • Check you are using the proper cables that do not exceed the recommended lengths • Verify that your switch and adapter port are compatible
Link light is on, but with no communication established	• Check that the latest driver is loaded • Check that both the adapter and its link are set to the same speed and duplex settings

Event message received of insufficient power	• When [adapter's current power consumption] > [PCIe slot advertised power limit] – a warning message appears in the server's system even logs (Eg. dmesg: "Detected insufficient power on the PCIe slow") • It's recommended to use a PCIe slot that can supply enough power. • If a message of the following format appears – "mlx5_core 0003:01:00.0: port_module:254:(pid 0): Port module event[error]: module 0, Cable error, One or more network ports have been powered down due to insufficient/unadvertised power on the PCIe slot" please upgrade your Adapter's firmware. • If the message remains – please consider switching from Active Optical Cable (AOC) or transceiver to Direct Attached Copper (DAC) connectivity.

Linux Troubleshooting

Environment Information	cat /etc/issue uname -acat /proc/cupinfo | grep 'model name' | uniqofed_info -sifconfig -aip link showethtool <interface>ethtool -i <interface_of_Mellanox_port_num>ibdev2netdev
Card Detection	lspci | grep -i Mellanox
Mellanox Firmware Tool (MFT)	Download and install MFT: MFT Documentation Refer to the User Manual for installation instructions.Once installed, run:mst startmst statusflint -d <mst_device> q
Ports Information	ibstat ibv_devinfo
Firmware Version Upgrade	To download the latest firmware version, refer to the NVIDIA Update and Query Utility.
Collect Log File	cat /var/log/messages dmesg >> system.logjournalctl (Applicable on new operating systems)cat /var/log/syslog

Windows Troubleshooting

Environment Information	From the Windows desktop choose the Start menu and run: `msinfo32` To export system information to a text file, choose the Export option from the File menu.Assign a file name and save.
Mellanox Firmware Tool (MFT)	Download and install MFT: MFT Documentation Refer to the User Manual for installation instructions.Once installed, open a CMD window and run:WinMFTmst startmst statusflint –d <mst_device> q
Ports Information	vstat
Firmware Version Upgrade	Download the latest firmware version using the PSID/board ID from here. flint –d <mst_device> –i <firmware_bin_file> b
Collect Log File	• Event log viewer • MST device logs: ◦ mst start ◦ mst status • flint –d <mst_device> dc > dump_configuration.log • mstdump <mst_device> dc > mstdump.log

Updating Adapter Firmware

Each adapter card is shipped with the latest version of qualified firmware at the time of manufacturing. However, NVIDIA issues firmware updates occasionally that provide new features and bug fixes. To check that your card is programmed with the latest available firmware version, download the mlxup firmware update and query utility. The utility can query for available NVIDIA adapters and indicate which adapters require a firmware update. If the user confirms, mlxup upgrades the firmware using embedded images. The latest mlxup executable and documentation are available in mlxup - Update and Query Utility .

```
[server1]# ./mlxup
Querying Mellanox devices firmware ...
Device Type:     ConnectX-6
Part Number:     MCX654106A-HCAT
Description:     ConnectX®-6 VPI adapter card, HDR IB (200Gb/s)
and 200GbE, dual-port QSFP56, Socket Direct 2x PCIe3.0 x16, tall
bracket
PSID:            MT_2190110032
PCI Device Name: 0000:06:00.0
Base GUID:       e41d2d0300fd8b8a
Versions:        Current            Available
                 FW 16.23.1020      16.24.1000

Status:          Update required

Device Type:     ConnectX-6
Part Number:     MCX654106A-HCAT
Description:     ConnectX®-6 VPI adapter card, HDR IB (200Gb/s)
and 200GbE, dual-port QSFP56, Socket Direct 2x PCIe3.0 x16, tall
bracket
PSID:            MT_2170110021
PCI Device Name: 0000:07:00.0
Base MAC:        0000e41d2da206d4
Versions:        Current          Available
                 FW 16.24.1000    16.24.1000

Status:          Up to date

Perform FW update? [y/N]: y
Device #1: Up to date
Device #2: Updating FW ... Done

Restart needed for updates to take effect.
```

```
Log File: /var/log/mlxup/mlxup-yyyymmdd.log
```

Monitoring

The adapter card incorporates the ConnectX IC, which operates in the range of temperatures between 0C and 105C.

There are three thermal threshold definitions that impact the overall system operation state:

- **Warning** – 105°C: On managed systems only: When the device crosses the 105°C threshold, a Warning Threshold message will be issued by the management SW, indicating to system administration that the card has crossed the Warning threshold. Note that this temperature threshold does not require nor lead to any action by hardware (such as adapter card shutdown).

- **Critical** – 115°C: When the device crosses this temperature, the firmware will automatically shut down the device.

- **Emergency** – 130°C: If the firmware fails to shut down the device upon crossing the Critical threshold, the device will auto-shutdown upon crossing the Emergency (130°C) threshold.

The card's thermal sensors can be read through the system's SMBus. The user can read these thermal sensors and adapt the system airflow in accordance with the readouts and the needs of the above-mentioned IC thermal requirements.

Specifications

MCX651105A-EDAT Specifications

> **Note**
>
> Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Physical	**Adapter Card Size**: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Single QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] **Auto-Negotiation**: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane), HDR (50Gb/s per lane) port		
	Ethernet: 200GBASE-CR4, 200GBASE-KR4, 200GBASE-SR4, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	InfiniBand	SDR/DDR/QDR/FDR/EDR/HDR100
		Ethernet	1/10/25/40/50/100 Gb/s
	PCI Express Gen3.0/4.0	SERDES @ 8.0GT/s/16GT/s, x8 lanes (2.0 and 1.1 compatible)	
	Voltage : 3.3Aux **Maximum current:** 100mA		

<table>
<tr><td rowspan="6">Power and Airflow</td><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">10.1W</td></tr>
<tr><td>Maximum Power</td><td colspan="4">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="5">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>TBD</td><td>TBD</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C[c]</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. The non-operational storage temperature specifications apply to the product without its package.

MCX653105A-HDAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Physical	**Adapter Card Size**: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Single QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] **Auto-Negotiation**: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane), HDR (50Gb/s per lane) port		
	Ethernet: 200GBASE-CR4, 200GBASE-KR4, 200GBASE-SR4, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	InfiniBand	SDR/DDR/QDR/FDR/EDR/HDR100/HDR
		Ethernet	1/10/25/40/50/100/200 Gb/s
	PCI Express Gen3/4:	SERDES @ 8.0GT/s/16GT/s, x16 lanes (2.0 and 1.1 compatible)	

<table>
<tr><td rowspan="8">Power and Airflow</td><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">19.3W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>350 LFM / 55°C</td><td>250 LFM / 35°C</td></tr>
<tr><td>NVIDIA Active 4.7W Cables</td><td>500 LFM / 55°C[c]</td><td>250 LFM / 35°C</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity[d]</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. For engineering samples - add 250LFM

d. The non-operational storage temperature specifications apply to the product without its package.

MCX653106A-HDAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Physical	**Adapter Card Size**: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Dual QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane), HDR (50Gb/s per lane) port		
	Ethernet: 200GBASE-CR4, 200GBASE-KR4, 200GBASE-SR4, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	InfiniBand	SDR/DDR/QDR/FDR/EDR/HDR100/HDR
		Ethernet	1/10/25/40/50/100/200 Gb/s
	PCI Express Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes (2.0 and 1.1 compatible)		

<table>
<tr><td rowspan="8">Power and Airflow</td><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">23.6W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>400 LFM / 55°C</td><td>300 LFM / 35°C</td></tr>
<tr><td>NVIDIA Active 4.7W Cables</td><td>950 LFM / 55°C
600 LFM / 48°Cd</td><td>300 LFM / 35°C</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity[c]</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td>Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td></td><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td></td><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. For both operational and non-operational states.

MCX653105A-HDAL Specifications

Note

Please make sure to install the ConnectX-6 card in an liquid-cooled Intel® Server System D50TNP platform.

Physical	**Adapter Card Size**: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Single QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane), HDR (50Gb/s per lane) port		
	Ethernet: 200GBASE-CR4, 200GBASE-KR4, 200GBASE-SR4, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	InfiniBand	SDR/DDR/QDR/FDR/EDR/HDR100/HDR
		Ethernet	1/10/25/40/50/100/200 Gb/s
	PCI Express Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes (2.0 and 1.1 compatible)		

<table>
<tr><td rowspan="8">Power and Airflow</td><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">18.5W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>TBD</td><td>TBD</td></tr>
<tr><td>NVIDIA Active 4.7W Cables</td><td>TBD</td><td>TBD</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity[c]</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. For both operational and non-operational states.

MCX653106A-HDAL Specifications

Note

Please make sure to install the ConnectX-6 card in an liquid-cooled Intel® Server System D50TNP platform.

Physical	**Adapter Card Size**: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Dual QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane), HDR (50Gb/s per lane) port		
	Ethernet: 200GBASE-CR4, 200GBASE-KR4, 200GBASE-SR4, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	InfiniBand	SDR/DDR/QDR/FDR/EDR/HDR100/HDR
		Ethernet	1/10/25/40/50/100/200 Gb/s
	PCI Express Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes (2.0 and 1.1 compatible)		

<table>
<tr><td rowspan="8">Power and Airflow</td><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">20.85W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="4">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>TBD</td><td>TBD</td></tr>
<tr><td>NVIDIA Active 4.7W Cables</td><td>TBD</td><td>TBD</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity[c]</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. For both operational and non-operational states.

MCX653105A-ECAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Physical	**Adapter Card Size**: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Single QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4 [a] Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane)		
	Ethernet: 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	**InfiniBand**	SDR/DDR/QDR/FDR/EDR/HDR100
		Ethernet	1/10/25/40/50/100 Gb/s
	PCIe Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes (2.0 and 1.1 compatible)		

<table>
<tr><td rowspan="8">Power and Airflow</td><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">15.6W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>300 LFM / 55°C</td><td>200 LFM / 35°C</td></tr>
<tr><td>NVIDIA Active 2.7W Cables</td><td>300 LFM / 55°C</td><td>200 LFM / 35°C</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C[c]</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. The non-operational storage temperature specifications apply to the product without its package.

MCX653106A-ECAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Note

For power specifications when using a single-port configuration, please refer to MCX653105A-ECAT Specifications

Physical	**Adapter** Card Size: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Dual QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] **Auto-Negotiation:** 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane) port		
	Ethernet: 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	**InfiniBand**	SDR/DDR/QDR/FDR/EDR
		Ethernet	1/10/25/40/50/100 Gb/s
	Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes (2.0 and 1.1 compatible)		

<table>
<tr><td rowspan="8">Power and Airflow</td><td colspan="4">Voltage: 12V, 3.3VAUX
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power^b</td><td>Passive Cables</td><td colspan="2">21.0W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>350 LFM / 55°C</td><td>250 LFM / 35°C</td></tr>
<tr><td>NVIDIA Active 2.7W Cables</td><td>550 LFM / 55°C</td><td>250 LFM / 35°C</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C^C</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. The non-operational storage temperature specifications apply to the product without its package.

MCX654105A-HCAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Physical	**Adapter Card Size:** 6.6 in. x 2.71 in. (167.65mm x 68.90mm) **Auxiliary PCIe Connection Card Size:** 5.09 in. x 2.32 in. (129.30mm x 59.00mm)Two Cabline CA-II Plus harnesses (white and black) Length: 35cm		
	Connector: Single QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] **Auto-Negotiation:** 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane), HDR (50Gb/s per lane) port		
	Ethernet: 200GBASE-CR4, 200GBASE-KR4, 200GBASE-SR4, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	**InfiniBand**	SDR/DDR/QDR/FDR/EDR/HDR100/HDR
		Ethernet	1/10/25/40/50/100/200 Gb/s
	Gen3: SERDES @ 8.0GT/s/, x16 lanes (2.0 and 1.1 compatible)		

<table>
<tr><td rowspan="5">Adapter Card Power</td><td colspan="4">Voltage: 12V, 3.3VAUX</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">27.1W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td colspan="5">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td>Active Auxiliary PCIe Connection Card Power</td><td>Typical Power</td><td colspan="3">3.0W</td></tr>
<tr><td></td><td>Maximum Power</td><td colspan="3">4.0W</td></tr>
<tr><td rowspan="3">Airflow</td><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>600 LFM / 55°C</td><td>350 LFM / 35°C</td></tr>
<tr><td>NVIDIA Active 4.7W Cables</td><td>600 LFM / 55°C[c]</td><td>350 LFM / 35°C</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°[c]</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C[d]</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. For engineering samples - add 250LFM

d. The non-operational storage temperature specifications apply to the product without its package.

MCX654106A-HCAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Note

For power specifications when using a single-port configuration, please refer to MCX654105A-HCAT Specifications

Physical	Low Profile Adapter Card Size: 6.6 in. x 2.71 in. (167.65mm x 68.90mm) Auxiliary PCIe Connection Card Size: 5.09 in. x 2.32 in. (129.30mm x 59.00mm)Two Cabline CA-II Plus harnesses (white and black) Length: 35cm
	Connector: Dual QSFP56 InfiniBand and Ethernet (copper and optical)

<table>
<tr><td rowspan="5">Protocol Support</td><td colspan="4">InfiniBand: IBTA v1.4[a]
Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane), HDR (50Gb/s per lane) port</td></tr>
<tr><td colspan="4">Ethernet: 200GBASE-CR4, 200GBASE-KR4, 200GBASE-SR4, 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR</td></tr>
<tr><td rowspan="2">Data Rate</td><td>InfiniBand</td><td colspan="2">SDR/DDR/QDR/FDR/EDR/HDR100/HDR</td></tr>
<tr><td>Ethernet</td><td colspan="2">1/10/25/40/50/100/200 Gb/s</td></tr>
<tr><td colspan="4">Gen3: SERDES @ 8.0GT/s, x16 lanes (2.0 and 1.1 compatible)</td></tr>
<tr><td rowspan="6">Adapter Card Power</td><td colspan="4">Voltage: 12V, 3.3VAUX</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">31.4W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td>Active Auxiliary PCIe Connection Card Power</td><td>Typical Power</td><td colspan="3">3.0W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">4.0W</td></tr>
<tr><td rowspan="3"></td><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>700 LFM / 55°C</td><td>400 LFM / 35°C</td></tr>
<tr><td>NVIDIA Active 4.7W Cables</td><td>1050 LFM / 55°C
600 LFM / 48°C</td><td>400 LFM / 35°C</td></tr>
</table>

Environmental	**Temperature**	**Operational**	0°C to 55°C
		Non-operational	-40°C to 70°C[c]
	Humidity	**Operational**	10% to 85% relative humidity
		Non-operational	10% to 90% relative humidity
	Altitude (Operational)	3050m	
Regulatory	**Safety**: CB / cTUVus / CE		
	EMC: CE / FCC / VCCI / ICES / RCM// KC		
	RoHS: RoHS Compliant		

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. The non-operational storage temperature specifications apply to the product without its package.

MCX654106A-ECAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Physical	Adapter Card Size: 6.6 in. x 2.71 in. (167.65mm x 68.90mm) Auxiliary PCIe Connection Card Size: 5.09 in. x 2.32 in. (129.30mm x 59.00mm)Two Cabline CA-II Plus harnesses (white and black) Length: 35cm
	Connector: Dual QSFP56 InfiniBand and Ethernet (copper and optical)

<table>
<tr><td rowspan="5">Protocol Support</td><td colspan="4">InfiniBand: IBTA v1.4[a]
Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane)</td></tr>
<tr><td colspan="4">Ethernet: 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR</td></tr>
<tr><td rowspan="2">Data Rate</td><td>InfiniBand</td><td colspan="2">SDR/DDR/QDR/FDR/HDR100/EDR</td></tr>
<tr><td>Ethernet</td><td colspan="2">1/10/25/40/50/100 Gb/s</td></tr>
<tr><td colspan="4">Gen3: SERDES @ 8.0GT/s, x16 lanes (2.0 and 1.1 compatible)</td></tr>
<tr><td rowspan="4">Adapter Card Power</td><td colspan="4">Voltage: 12V, 3.3VAUX
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">27.1W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="2">Active Auxiliary PCIe Connection Card Power</td><td>Typical Power</td><td colspan="3">3.0W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">4.0W</td></tr>
<tr><td rowspan="3">Airflow</td><td rowspan="3">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>600 LFM / 55°C</td><td>400 LFM / 35°C</td></tr>
<tr><td></td><td>NVIDIA Active 2.7W Cables</td><td>700 LFM / 55°C</td><td>400 LFM / 35°C</td></tr>
</table>

Environmental	**Temperature**	**Operational**	0°C to 55°C
		Non-operational	-40°C to 70°C[c]
	Humidity	**Operational**	10% to 85% relative humidity
		Non-operational	10% to 90% relative humidity
	Altitude (Operational)	3050m	
Regulatory	**Safety**: CB / cTUVus / CE		
	EMC: CE / FCC / VCCI / ICES / RCM / KC		
	RoHS: RoHS Compliant		

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. The non-operational storage temperature specifications apply to the product without its package.

MCX653105A-EFAT Specifications

> **Note**
>
> Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Physical	**Adapter C**ard Size: 6.6 in. x 2.71 in. (167.65mm x 68.90mm)
	Connector: Single QSFP56 InfiniBand and Ethernet (copper and optical)

<table>
<tr><td rowspan="5">Protocol Support</td><td colspan="3">InfiniBand: IBTA v1.4[a]
Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane) port</td></tr>
<tr><td colspan="3">Ethernet: 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR</td></tr>
<tr><td rowspan="2">Data Rate</td><td>InfiniBand</td><td>SDR/DDR/QDR/FDR/EDR/HDR100</td></tr>
<tr><td>Ethernet</td><td>1/10/25/40/50/100 Gb/s</td></tr>
<tr><td colspan="3">Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes, Socket Direct (2.0 and 1.1 compatible)</td></tr>
<tr><td rowspan="5">Adapter Card Power</td><td colspan="3">Voltage: 12V, 3.3VAUX
Maximum current: 100mA</td></tr>
<tr><td>Power</td><td colspan="2">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td>19.4W</td></tr>
<tr><td>Maximum Power</td><td colspan="2">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="3">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="4">Airflow</td><td rowspan="4">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>300 / 55°C</td><td>200 / 35°C</td></tr>
<tr><td>NVIDIA Active 2.75W Cables</td><td>300 / 55°C</td><td>200 / 35°C</td></tr>
</table>

Environmental	**Temperature**	**Operational**	0°C to 55°C
		Non-operational	-40°C to 70°C[c]
	Humidity	**Operational**	10% to 85% relative humidity
		Non-operational	10% to 90% relative humidity[c]
	Altitude (Operational)	3050m	
Regulatory	**Safety**: CB / cTUVus / CE		
	EMC: CE / FCC / VCCI / ICES / RCM / KC		
	RoHS: RoHS Compliant		

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. The non-operational storage temperature specifications apply to the product without its package.

MCX653106A-EFAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

Note

For power specifications when using a single-port configuration, please refer to MCX653105A-EFAT Specifications.

Physical	**Adapter Card Size:** 6.6 in. x 2.71 in. (167.65mm x 68.90mm)		
	Connector: Dual QSFP56 InfiniBand and Ethernet (copper and optical)		
Protocol Support	**InfiniBand**: IBTA v1.4[a] Auto-Negotiation: 1X/2X/4X SDR (2.5Gb/s per lane), DDR (5Gb/s per lane), QDR (10Gb/s per lane), FDR10 (10.3125Gb/s per lane), FDR (14.0625Gb/s per lane), EDR (25Gb/s per lane) port, HDR100 (2 lane x 50Gb/s per lane) port		
	Ethernet: 100GBASE-CR4, 100GBASE-CR2, 100GBASE-KR4, 100GBASE-SR4, 50GBASE-R2, 50GBASE-R4, 40GBASE-CR4, 40GBASE-KR4, 40GBASE-SR4, 40GBASE-LR4, 40GBASE-ER4, 40GBASE-R2, 25GBASE-R, 20GBASE-KR2, 10GBASE-LR,10GBASE-ER, 10GBASE-CX4, 10GBASE-CR, 10GBASE-KR, SGMII, 1000BASE-CX, 1000BASE-KX, 10GBASE-SR		
	Data Rate	InfiniBand	SDR/DDR/QDR/FDR/EDR, HDR100
		Ethernet	1/10/25/40/50/100 Gb/s
	Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes, Socket Direct (2.0 and 1.1 compatible)		

<table>
<tr><td rowspan="9">Adapter Card Power and Airflow</td><td colspan="4">Voltage: 12V, 3.3VAUX</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power^b</td><td>Passive Cables</td><td colspan="2">21.6W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td colspan="4">Maximum power available through QSFP56 port: 5W</td></tr>
<tr><td rowspan="4">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>350 LFM / 55°C</td><td>250 LFM / 35°C</td></tr>
<tr><td>NVIDIA Active 2.75W Cables</td><td>550 LFM / 55°C</td><td>250 LFM / 35°C</td></tr>
<tr><td rowspan="5">Environmental</td><td rowspan="2">Temperature</td><td>Operational</td><td colspan="2">0°C to 55°C</td></tr>
<tr><td>Non-operational</td><td colspan="2">-40°C to 70°C^c</td></tr>
<tr><td rowspan="2">Humidity</td><td>Operational</td><td colspan="2">10% to 85% relative humidity</td></tr>
<tr><td>Non-operational</td><td colspan="2">10% to 90% relative humidity</td></tr>
<tr><td>Altitude (Operational)</td><td colspan="3">3050m</td></tr>
<tr><td rowspan="3">Regulatory</td><td colspan="4">Safety: CB / cTUVus / CE</td></tr>
<tr><td colspan="4">EMC: CE / FCC / VCCI / ICES / RCM / KC</td></tr>
<tr><td colspan="4">RoHS: RoHS Compliant</td></tr>
</table>

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

b. Typical power for ATIS traffic load.

c. The non-operational storage temperature specifications apply to the product without its package.

MCX683105AN-HDAT Specifications

Note

Please make sure to install the ConnectX-6 card in a PCIe slot that is capable of supplying the required power and airflow as stated in the below table.

<table>
<tr><td rowspan="2">Physical</td><td colspan="4">Adapter Card Size: 5.59 in. x 2.71 in. (142mm x 68.90mm)</td></tr>
<tr><td colspan="4">Connector: Single QSFP InfiniBand (copper and optical)</td></tr>
<tr><td rowspan="3">Protocol Support</td><td colspan="4">InfiniBand: IBTA v1.4[a]
HDR (4 lanes x 50Gb/s per lane) , HDR100 (2 lane x 50Gb/s per lane), EDR (4 lanes x 25Gb/s per lane), FDR (4 lanes x 14.0625Gb/s), 1X/2X/4X SDR (4 lanes x 2.5Gb/s per lane).</td></tr>
<tr><td>Data Rate</td><td>InfiniBand</td><td colspan="2">HDR/HDR100/EDR/FDR/SDR</td></tr>
<tr><td colspan="4">Gen3/4: SERDES @ 8.0GT/s/16GT/s, x16 lanes</td></tr>
<tr><td rowspan="5">Adapter Card Power</td><td colspan="4">Voltage: 12V, 3.3VAUX</td></tr>
<tr><td>Power</td><td colspan="3">Cable</td></tr>
<tr><td>Typical Power[b]</td><td>Passive Cables</td><td colspan="2">19.58W</td></tr>
<tr><td>Maximum Power</td><td colspan="3">Please refer to ConnectX-6 VPI Power Specifications (requires NVONline login credentials)</td></tr>
<tr><td colspan="4">Voltage : 3.3Aux
Maximum current: 100mA</td></tr>
<tr><td colspan="5">Maximum power available through QSFP port: 5W</td></tr>
<tr><td rowspan="4">Airflow</td><td rowspan="4">Airflow (LFM) / Ambient Temperature</td><td rowspan="2">Cable Type</td><td colspan="2">Airflow Direction</td></tr>
<tr><td>Heatsink to Port</td><td>Port to Heatsink</td></tr>
<tr><td>Passive Cables</td><td>TBD</td><td>TBD</td></tr>
<tr><td>NVIDIA Active 2.75W Cables</td><td>TBD</td><td>TBD</td></tr>
</table>

Environmental	**Temperature**	**Operational**	0°C to 55°C
		Non-operational	-40°C to 70°C[c]
	Humidity	**Operational**	10% to 85% relative humidity
		Non-operational	10% to 90% relative humidity
	Altitude (Operational)	3050m	
Regulatory	**Safety**: CB / cTUVus / CE		
	EMC: CE / FCC / VCCI / ICES / RCM / KC		
	RoHS: RoHS Compliant		

Notes: a. The ConnectX-6 adapters supplement the IBTA auto-negotiation specification to get better bit error rates and longer cable reaches. This supplemental feature only initiates when connected to another NVIDIA InfiniBand product.

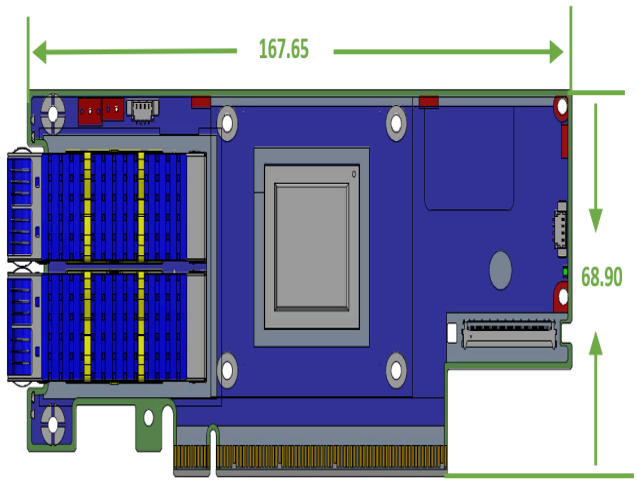
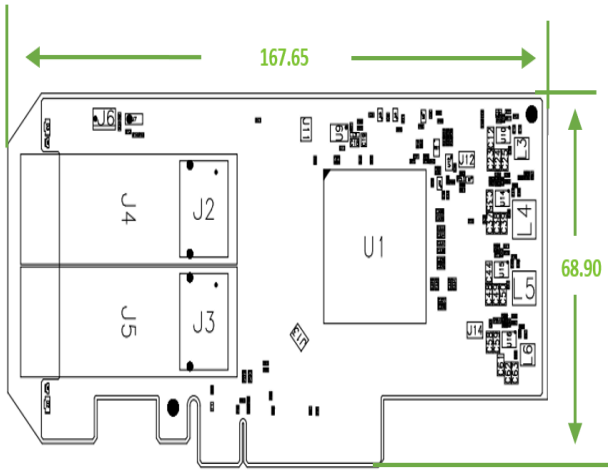
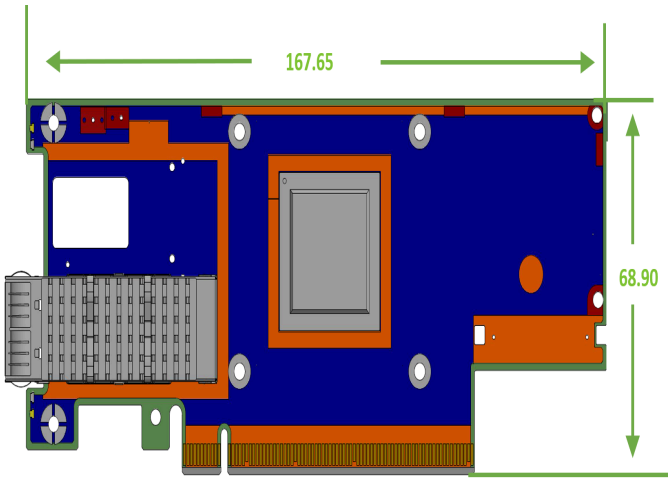
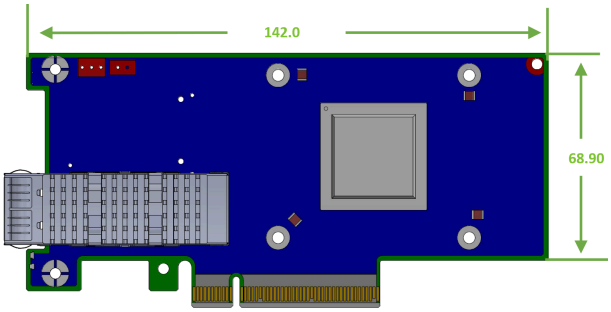
b. Typical power for ATIS traffic load.

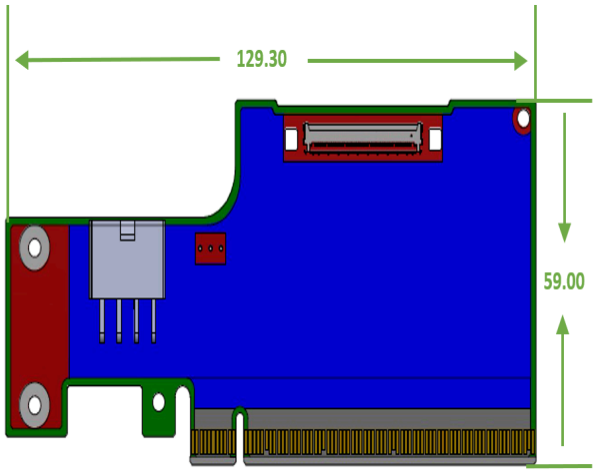
c. The non-operational storage temperature specifications apply to the product without its package.

Adapter Card and Bracket Mechanical Drawings and Dimensions

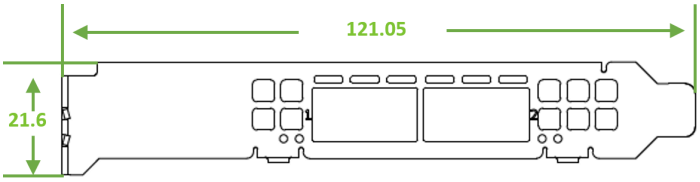
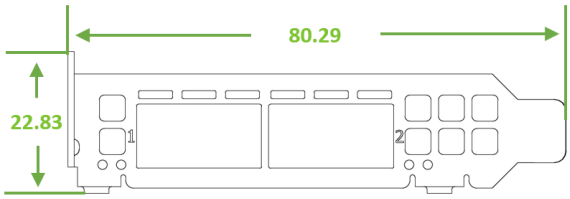
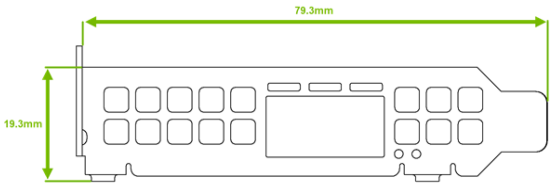
> **Note**
>
> All dimensions are in millimeters. The PCB mechanical tolerance is +/- 0.13mm.

Adapter Cards

ConnectX-6 PCIe x16 Dual-port Adapter Card	ConnectX-6 PCIe x8 Dual-port Adapter Card
	
ConnectX-6 PCIe x16 Single-port Adapter Card	ConnectX-6 PCIe x8 Single-port Adapter Card
	
MCX653105A-HDAL	Auxiliary PCIe Connection Card

ConnectX-6 PCIe x16 Dual-port Adapter Card	ConnectX-6 PCIe x8 Dual-port Adapter Card
	

Brackets Dimensions

Dual-port Adapter Tall Bracket	Dual-port Adapter Short Bracket
	
Single-port Adapter Tall Bracket	Single-port Adapter Short Bracket
	

PCI Express Pinouts Description for Single-Slot Socket Direct Card

Note

This section applies to ConnectX-6 single-slot cards (MCX653105A-EFAT and MCX653106A-EFAT).

ConnectX-6 single-slot Socket Direct cards offer improved performance to dual-socket servers by enabling direct access from each CPU in a dual-socket server to the network through its dedicated PCIe interface. The PCIe x16 interface is split into two PCIe x8 in a row, such that each of the PCIe x8 lanes can be connected to a dedicated CPU in a dual-socket server. In such a configuration, Socket Direct brings lower latency and lower CPU utilization as the direct connection from each CPU to the network means the Interconnect can bypass a QPI (UPI) and the other CPU, optimizing performance and improving latency. CPU utilization is improved as each CPU handles only its own traffic and not traffic from the other CPU.

In order to allow this capability, a system with a special PCI Express x16 slot is required. Table 31 provides the pin definitions of the required four special PCIe pins.

PCIe pin#	Server Connection for 2x PCIe x8 mode	Server Connection for 1x PCIe x16 mode	Comments
B82	1Kohm pull up to 3.3VAUX	Either leave unconnected or pulled to GND	Configure the card to work at 2x PCIe x8 or 1x PCIe x16 modes

PCIe pin#	Server Connection for 2x PCIe x8 mode	Server Connection for 1x PCIe x16 mode	Comments
A32	P signal of differential PCIe clock (100MHz nominally) of the CPU which connects to PCIe lanes 15-8 of the PCIe connector	Not Connected	PCIe clock for ConnectX-6 PCIe bus lanes [15:8]
A33	N signal of differential PCIe clock (100MHz nominally) of the CPU which connects to PCIe lanes 15-8 of the PCIe connector	Not Connected	
A50	PERST signal from the CPU which connects to PCIe lanes 15-8 of the PCIe connector	Not Connected	PERST (PCIe Reset) for ConnectX-6 PCIe bus lanes [15:8]

Finding the GUID/MAC on the Adapter Card

Each NVIDIA adapter card has a different identifier printed on the label: serial number and the card MAC for the Ethernet protocol and the card GUID for the InfiniBand protocol. VPI cards have both a GUID and a MAC (derived from the GUID).

> **Note**
>
> The product revisions indicated on the labels in the following figures do not necessarily represent the latest revisions of the cards.



MCX683105AN-HDAT Board Label (Example)

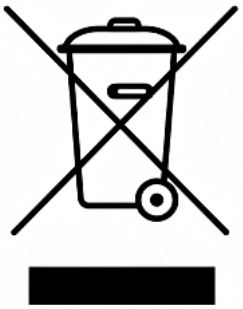

NVIDIA
ConnectX-6 DE HDR
Model: CX683105A
P/N: MCX683105AN-HDAT
S/N: MT0806X01504
GUID: 0002C90300270510
Made in Israel / Rev:A2 / 2022-07-24

Compliance

The NVIDIA ConnectX is compliant with the regulations listed in this section.

Country	Compliance	Compliance Mark
United States	Federal Communications Commission (FCC) FCC Marking (Class A) This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including any interference that may cause undesired operation of the device. NOTE: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense. California Department of Toxic Substances Control: Perchlorate Material - special handling may apply. Refer to www.dtsc.ca.gov/perchlorate.	

Country	Compliance	Compliance Mark
United States/Canada	TÜV Rheinland of North America is accredited as a Nationally Recognized Testing Laboratory (NRTL), by OSHA (The Occupational Safety and Health Administration) in the United States, and as a Product Certification Body by SCC (Standards Council of Canada) in Canada. Refer to https://www.tuv.com/usa/en/ctuvus-certification.html	
Canada	Innovation, Science and Economic Development Canada (ISED) CAN ICES-3(A)/NMB-3(A) The Class A digital apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulation. Cet appareil numerique de la class A respecte toutes les exigences du Reglement sur le materiel brouilleur du Canada.	

Country	Compliance	Compliance Mark
CE	European Conformity; Conformité Européenne (CE) This is a Class A product. In a domestic environment this product may cause radio frequency interference in which case the user may be required to take adequate measures. This device bears the CE mark in accordance with Directive 2014/53/EU. This device complies with the following Directives: • EMC Directive A, I.T.E Equipment. • Low Voltage Directive for electrical safety. • RoHS Directive for hazardous substances. • Energy-related Products Directive (ErP). The full text of EU declaration of conformity is available at the following URL: http://www.nvidia.com/support A copy of the Declaration of Conformity to the essential requirements may be obtained directly from NVIDIA GmbH (Bavaria Towers – Blue Tower, Einsteinstrasse 172, D-81677 Munich, Germany).	 
Australia and New Zealand	Australian Communications and Media Authority This product meets the applicable EMC requirements for Class A, I.T.E equipment.	
Brazil	**INMETRO**	

Country	Compliance	Compliance Mark
Japan	**Voluntary Control Council for Interference (VCCI)** This is a Class A product. In a domestic environment this product may cause radio interference, in which case the user may be required to take corrective actions. VCCI-A. 2008年、日本における□品含有表示方法、JISC0950が公示されました。□造事□者は、2006年7月1日以降に□□される□□□□子□器の特定化学物□の含有に付きまして情□提供を□□付けられました。 □品の部材表示に付きましては、以下をご□ください。 A Japanese regulatory requirement, defined by specification JIS C 0950, 2008, mandates that manufacturers provide Material Content Declarations for certain categories of electronic products offered for sale after July 1, 2006.	 この装置は、クラス 用すると□波妨害を引 合には使用者が□切な あります。

<table>
<tr><th>Country</th><th>Compliance</th><th>Compliance Mark</th></tr>
<tr><td>Japan</td><td>Japan RoHS Material Content Declaration
日本工□□格 JIS C 0950:2008により、2006年 7月 1日以降に□□される特定分野の□□および□子□器について、□造者による含有物□の表示が□□付けられます。□器名称：リバ

<table>
<tr><th>主な分□</th><th>Pb</th><th>Hg</th><th>Cd</th><th>Cr(VI)</th><th>PBB</th><th>PBDE</th></tr>
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<tr><td>プリント基板</td><td>除外□目</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>プロセッサ□</td><td>除外□目</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>マザ□ボ□ド</td><td>除外□目</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>□源</td><td>除外□目</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>システムメモリ</td><td>除外□目</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
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</td><td>Material Content D

A Japanese regulat
specification JIS C
manufacturers prov
for certain categori
for sale after July 1
Product Model Nur

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<tr><th>Major Classification</th><th>Pb</th></tr>
<tr><td>Chassis</td><td>Exe</td></tr>
<tr><td>PCA</td><td>Exe</td></tr>
<tr><td>Processor</td><td>Exe</td></tr>
<tr><td>Motherboard</td><td>Exe</td></tr>
<tr><td>Power supply</td><td>Exe</td></tr>
<tr><td>System Memory</td><td>Exe</td></tr>
<tr><td>Hard Disk Drive</td><td>Exe</td></tr>
<tr><td>Mechanical Parts (Fans, Heatsinks, Bezels..)</td><td>Exe</td></tr>
<tr><td>Cables / Connectors</td><td>Exe</td></tr>
<tr><td>Soldering Material</td><td>0</td></tr>
<tr><td>Flux, Solder Paste, Labels, Consumables</td><td>0</td></tr>
</table>
</td></tr>
</table>

Country	Compliance							Compliance Mark
	ハードディスクドライブ	除外項目	0	0	0	0	0	**Major Classification** **Pb** **Notes:** • **"0"** indicates chemical sub level specifie 2008. • **"Exempt"** ind chemical sub and it is not r for that spec standard, JIS • **"Exceeding 0** is entered in specified che threshold lev 0950: 2008.
	機械部品（ファン、ヒートシンク、ベゼル..）	除外項目	0	0	0	0	0	
	ケーブル / コネクター	除外項目	0	0	0	0	0	
	はんだ付け材料	0	0	0	0	0	0	

Country	Compliance							Compliance Mark
	フラックス、クリ□ムはんだ、ラベル、その他消耗品	0	0	0	0	0	0	

1. 0」は、特定化学物□の含有率が日本工□□格 JIS C 0950:2008 に□□されている含有率基□□より低いことを示します。
2. 「除外□目」は、特定化学物□が含有マ□クの除外□目に□当するため、特定化学物□について、日本工□□格 JIS C 0950:2008 に基づく含有マ□クの表示が不要であることを示します。
3. 「0.1wt%超」または「0.01wt%超」は、特定化学物□の含有率が日本工□□格 JIS C 0950:2008 に□□されている含有率基□□を超えていることを示します。

Country	Compliance	Compliance Mark
South Korea	Korean Agency for Technology and Standards (KATS) Class A Equipment (Industrial Broadcasting & Communication Equipment). This equipment Industrial (Class A) electromagnetic wave suitability equipment and seller or user should take notice of it, and this equipment is to be used in the places except for home. Korea RoHS Material Content Declaration	**R-R-WT1-P3687** **A□ □□ (□□□ □□□ □□□□)** **□ □□□ □□□ □□□ □□□, □**

확인 및 평가 양식은 제품에 포함 된 유해 물질의 허용 기준의 준수에 관한

문 준비	상호 :	엔비디아홍콩홀딩즈 리미티드(영업소)	법인등록번호	110181-0036373
	대표자성명	카렌테레사번즈	사업자등록번호:	120-84-06711
	주소	서울특별시 강남구 영동대로 511, 2101호 (삼성동,		

제품 내용

제품의 종류	해당없음	제품명(규격)	해당없음
세부모델명(변호)	해당없음	제품출시일	해당없음
제품의 중량	해당없음	제조, 수입업자	엔비디아

엔비디아의 그래픽 카드제품은 전기 전자제품 및 자동차의 자원순환에 관한 법률 시행령 제 11조 제 1항에 의거한 법 시행령규칙 제 3조에에따른 유해물질함유 기준을 확인 및 평가한 결과, 이를 준수하였음을 공표합니다.

구비서류 : 없음

작성방법

① 제품의 종류는 "전기.전자제품 및 자동차의 자원순환에관한 법률 시행령" 제 8조 제 1항 및 제 2항에 따른 품목별로 구분하여 기재합니다.

② 전기 전자 제품의 경우 모델명 (번호), 자동차의 경우, 제원관리번호를 기재합니다.

③ 해당제품의 제조업자 또는 수입업자를 기재합니다.

Confirmation and Evaluation Form Concerning the Adherence to Acceptable Standards of Hazardous Materials Contained in Products

Statement Prepared by	Company Name:	Nvidia HongKong Holding Ltd.Korea branch	Corporate Identification Number:	110181-0036373
	Name of Company Representative:	Karen Theresa Burns	Business Registration Number:	120-84-06711
	Address:	2788 San Tomas Expressway, Santa Clara, CA 95051		

Product Information

Product Category:	N/A	Name of Product:	N/A
Detailed Product Model Name (Number):	N/A	Date of first market release:	N/A
Weight of Product:	N/A	Manufacturer and/or Importer:	NVIDIA Corporation

This for is publicly certify That NVIDIA Company has undergone the confirmation and evaluation procedures for the acceptable amounts of hazardous materials contained in graphic card according to the regulations stipulated in Article 3 of the 'Status on the Recycling of Electrical and Electronic Products, and Automobiles' and that company has graphic card adhered to the Enforcement Regulations of Article 11, Item 1 of the statute.

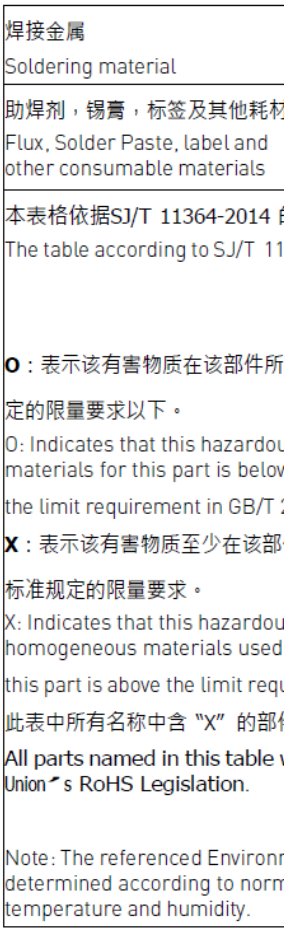
Attachment: None

*Preparing the Form

① Please indicate the product category according to the categories listed in Article 8, Items 1and 2 of the " Enforcement Ordinance of the Statute on the Recycling of Electrical, Electronic and Automobile Materials"

② For electrical and electronic products, please indicate the Model Name (and number). For automobiles, please indicate the Vehicle Identification Number.

③ Please indicate the name of manufacturer and/or importer of the product.

Country	Compliance	Compliance Mark
China	China Compulsory Certificate No certification is needed for China. The NVIDIA DGX H100/H200 system is a server with power consumption greater than 1.3 kW. China RoHS Material Content Declaration	

Country	Compliance	Compliance Mark
Taiwan	Bureau of Standards, Metrology & Inspection (BSMI)	R33088 RoHS 報驗義務人： 香港商輝達香港控股有 臺北市內湖區基湖路8
	Taiwan RoHS Material Content Declaration	Declaration of the 單元 Parts 機箱 Chassis 印刷電路部件 PCA 處理器 Processor 主板 Motherboard 電源設備 Power supply 存儲設備 System memory 硬盤驅動器 Hard drive 機械部件（風扇、散熱器、面板等） Mechanical parts (fan, heat sink, bezel...) 線材/連接器 Cables/Connectors 焊接金屬 Soldering material 助焊劑，錫膏，標籤及其他耗材 Flux, Solder Paste, label and other consumable materials

Country	Compliance	Compliance Mark
Russia/Kazakhstan/Belarus	Customs Union Technical Regulations (CU TR) This device complies with the technical regulations of the Customs Union (CU TR) ТЕХНИЧЕСКИЙ РЕГЛАМЕНТ ТАМОЖЕННОГО СОЮЗА О безопасности низковольтного оборудования (ТР ТС 004/2011) ТЕХНИЧЕСКИЙ РЕГЛАМЕНТ ТАМОЖЕННОГО СОЮЗА Электромагнитная совместимость технических средств (ТР ТС 020/2011) Технический регламент Евразийского экономического союза “Об ограничении применения опасных веществ в изделиях электротехники и радиоэлектроники” (ТР ЕАЭС 037/2016) Federal Agency of communication (FAC) This device complies with the rules set forth by Federal Agency of Communications and the Ministry of Communications and Mass Media. Federal Security Service notification has been filed.	
Israel	ודא שלמות ותקינות כבל החשמל והתקע אין להכניס או להוציא את התקע מרשת החשמל בידיים רטובות . אין לפתוח את המכשיר , במקרה של בעיה כלשהו יש לפנות למעבדת השירות הקרובה. יש להרחיק את המכשיר ממנזלים . במקרה של ריח מוזר, רעשים שמקורם במכשיר , יש לנתקו מיידית מרשת החשמל ולפנות למעבדת שירות המכשיר מיועד לשימוש בתוך המבנה , ולא לשימוש חיצוני ולא לשימוש בסביבה לחה. אין לחתוך, לשבור, ולעקם את הכבל החשמל. אין להניח חפצים על הכבל החשמל או להניח לו להתחמם יתר על המידה , שכן עלול לגרום לנזק ,דליקה או התחשמלות . יש להקפיד לחזק את התקן הניתוק במצב תפעולי מוכן לשימוש. אזהרה: אין להחליף את כבל הזינה בתחליפים לא מקוריים, חיבור לקוי עלול לגרום להתחשמלות המשתמש. בשימוש על כבל מאריך יש לוודא תקינות מליך הארקה שבכבל .	

Country	Compliance	Compliance Mark
India	Bureau of India Standards (BIS) Authenticity may be verified by visiting the Bureau of Indian Standards website at http://www.bis.gov.in. India RoHS Compliance Statement This product, as well as its related consumables and spares, complies with the reduction in hazardous substances provisions of the “India E-waste (Management and Handling) Rule 2016”. It does not contain lead, mercury, hexavalent chromium, polybrominated biphenyls or polybrominated diphenyl ethers in concentrations exceeding 0.1 weight % and 0.01 weight % for cadmium, except for where allowed pursuant to the exemptions set in Schedule 2 of the Rule.	
South Africa	South African Bureau of Standards (SABS) This device complies with the following SABS Standards: SANS 2332: 2017/CISPR 32:2015 SANS 2335:2018/ CISPR 35:2016 National Regulator of Compulsory Specification (NRCS) This device complies with following standard under VC 8055: SANS IEC 60950-1	

Country	Compliance	Compliance Mark
Great Britain (England, Wales, and Scotland)	UK Conformity Assessed This device complies with the following Regulations: • SI 2016/1091: Electromagnetic Compatibility (EMC) • SI 2016/1101: The Low Voltage Electrical Equipment (Safety) • SI 2012/3032: The Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment (As Amended) A copy of the Declaration of Conformity to the essential requirements may be obtained directly from NVIDIA Ltd. (100 Brook Drive, 3rd Floor Green Park, Reading RG2 6UJ, United Kingdom)	UKCA

Document Revision History

Date	Comments/Changes
Mar. 2026	Added Compliance
Nov. 2023	Added the 100GBASE-CR2 protocol to Specifications tables
Aug. 2023	Updated bracket dimensions in Specifications
May. 2023	• Updated Specifications to include non-operational storage temperature specifications • Updated dual-port bracket dimensions
Apr. 2023	Updated Monitoring
Mar. 2023	Updated mechanical drawings in Specifications
Dec. 2022	• Updated mechanical drawings in Specifications • Updated label in Finding the GUID/MAC on the Adapter Card
Aug. 2022	• Added mechanical drawing and board label example for MCX683105AN-HDAT. • Added a note about FRU EEPROM memory component under the Features and Benefits table.
Jul. 2022	Updated the "SMBus Interface" under Interfaces.
Jan. 2022	Added MCX683105AN-HDAT.
Mar. 2021	Updated "Troubleshooting".
Dec. 2020	Updated installation instructions.

Date	Comments/Changes
Dec. 2020	Added MCX653105A-HDAL and MCX653106A-HDAL support across the document.
Mar. 2020	Added MCX651105A-EDAT support across the document.
Sep. 2019	Added a note to the hardware installation instructions.
Aug. 2019	Updated "Package Contents" and "Hardware Installation"
Aug. 2019	Updated "PCI Express Pinouts Description".
Aug. 2019	Updated "Hardware Installation".
Jul. 2019	Updated "Linux Driver" and "Identifying the card in the system" to include lspci command output examples.
Jun. 2019	• Added MCX653105A-HDAT and MCX654105A-HCAT to the UM. • Updated "LED Interfaces".
Jun. 2019	• Added a note to "Windows Driver Installation". • Added short and tall brackets dimensions.
May. 20.19	• Added mechanical drawings to "Specifications". • Updated PCB mechanical tolerance in "Specifications".
May. 2019	• Updated "LEDs Interface" specifications. • Updated PCB mechanical tolerance in "Specifications".
Apr. 2019	• Migrated to on-line format; minor reorganization. • Added a note to "Introduction"
Feb. 2019	Updated “Specifications"
Feb. 2019	Updated “Specifications”

Date	Comments/Changes
Jan. 2019	• Updated “Airflow Specifications” • Added a note to “Installation Instructions”
Dec. 2018	Updated “Airflow Specifications”
Dec. 2018	• Updated “Hardware Requirements” • Updated “Product Overview”
Nov. 2018	Updated “Hardware Requirements”
Oct. 2018	First release

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